

**GIS-Based Analysis of Commuters Facility and Modern Jeepney Routes
Using Contingency Valuation and K-means Clustering
along Barangay Gusa to Barangay Bugo in Cagayan de Oro City**

A Project Study Presented to
The Faculty of the Department of Civil Engineering
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CHAPTER ONE

THE PROBLEM AND ITS SETTING

1.1 Background of the Study

The Philippine public transportation is currently experiencing significant transitions and experiences. These systemic challenges include urban congestion, operational consistency, and service reliability (Francisco, 2025). One of the challenges experienced in the Philippines in regards to commuting is the lack of unified transport data. This contributes to the hindrance of implementation of modernization policies locally (NCTS, 2024). This results in the necessity of transformative reforms to uplift the commuter experience. In 2017, the government launched the Public Utility Vehicle Modernization Program (PUVMP). With the intention not just to change physical aspects of the fleet but also to improve the operations and introduce technologies that are necessary for commuter safety and public acceptance (Gumasing et al., 2024). The implementation shows success as an evaluation conducted by the German Agency for International Cooperation (GIZ, 2019) indicates that there are significant performance improvements in operation. The operation on modernized vehicles achieved 19 vehicle performing hours versus 14 hours for the traditional routes. This data underscores the national strategic shift toward data-driven, quality-based public transport assessment. But despite these improvements in transportation by equipping modern jeepneys with GPS- telematics systems, a lack of tools for monitoring to assess service improvements. (Cunanan, 2025)

Cagayan de Oro city (CDO), a highly urbanized and a gateway city in Northern Mindanao. The city is designated as a key area for the implementation of PUVMP (Philippine News Agency, 2024). The redefinition starts off with the vehicle type with new, Euro-4 compliant, air-conditioned units (Bureau of Philippine standards, 2017). The long-term success of the implementation of PUVMP in the CDO does not just solely rely on the physical attributes, and facilities of the PUVMP but also critically relies on the operational advantages and upgrading the supporting infrastructure, particularly within the six selected barangays that experience the highest passenger and traffic volumes. This technological upgrade's full potential is hampered by two core gaps. First is the empirical operational advantages of the new fleet; in terms of factors like schedule reliability and travel speed, but remains unverified due to a lack of public, quantitative data derived from the integrated GPS/AFCs technologies. As studied by Reyes et al. (2023), there is a need for change in methodology towards utilizing GPS data progress from subjective observation to objective, engineering based indicators, without these data the operational metrics remain unmeasured. to Second, the necessary support infrastructure, including designated passenger waiting areas and waiting facilities, has not kept pace with vehicle modernization (Agaton, et al., 2023). This infrastructural deficiency contributes to operational stagnation and hinders the holistic improvement of the commuter experience (Mayo & Taboada, 2021).

To support the sustainable adaptation of the PUVMP in CDO, a quantitative and spatial assessment is required. This study addresses the gaps by conducting a Geospatial Analysis of the Modern Jeepney system's performance. By providing evidence-based, and technical data; this research aims to assist community stakeholders and local governance. To foresee which locations are a priority in developing the placement of passenger facilities - waiting sheds. This study aims to suggest which locations to uphold and erect passenger facilities not just for the communities but to also refine the overall public transport framework.

1.2 Statement of the Problem

The ongoing urbanization and economic growth of Cagayan de Oro City have raised the demand for reliable public transportation facilities, especially the modernized Public Utility Jeepneys (PUJs), which plays a crucial role in the daily lives of residents of the 6 Selected Barangays. These Jeepneys connect people to key destinations like workplaces and markets, yet the system's operational efficiency is often undermined by insufficient support infrastructure and outdated planning approaches. Problems such as chaotic boarding and alighting procedures, commuters enduring extreme weather without shelter, and persistent traffic bottlenecks not only diminish road performances but also jeopardize passenger safety and exacerbate congestion. These challenges are further compounded by a scarcity of in-depth, data-informed research that examines passenger needs, movement trends, route viability, and spatial reach throughout the urban barangay landscape.

Moreover, the lack of passenger facilities like waiting sheds leaves commuters without adequate protection, comfort. Existing routes and setups are typically shaped by casual observations rather than just systematic analysis, resulting in uneven service distribution and inefficiencies that affect everyone's daily routines. This highlights the urgent need for rigorous studies that delve into the working of modern PUJs and offer practical recommendations to strengthen infrastructure, making public transport more accessible, secure, and environmentally conscious.

This study will evaluate the operational efficiency and supporting structures of the modern jeepney system in the 6 Selected Barangays of Cagayan de Oro City. Specifically to address the following research questions, aligned with the study's objectives:

1. What are the baseline conditions of the selected barangays in terms of existing jeepney routes, road corridors, and commuter infrastructure within the study area?
2. What are the commuters' perceptions of jeepney routes, trips, and services in terms of fare, travel distance, and delays, and what is their willingness to pay (WTP) for improved passenger facilities using the contingent valuation (CV) approach?
3. What spatial patterns and cluster of potential passenger facility locations can be identified using K-means clustering analysis
4. How can GIS-based spatial mapping be used to identify and visualize recommended locations for sheds to support transport planning and decision making?

1.3 Objectives of the Study

The main objective of this study is to develop geospatial analysis of commuter facility and prioritization of sites for passenger facilities of Modern Jeepneys in selected barangays in Cagayan de Oro City. Moreover, the study will address the following specific objectives, namely:

1. To determine baseline data of the selected barangay, the jeepney routes, and the road corridor.
2. To conduct commuters' perception surveys on jeepney routes, trips, and services (fare, distance, and delay) using CV and WTP.
3. To determine spatial patterns and clustering of potential facility locations using k-means analysis.
4. To develop GIS-based maps on recommended jeepney terminals, loading and unloading areas, and waiting stations.

1.4 Significance of the Study

This study is important because it examines the commuter facilities and jeepney routes from Barangay Gusa to Barangay Bugo in Cagayan de Oro City. By using GIS mapping, contingency valuation, and K-means clustering, the study identifies areas where commuters usually wait, load, and unload. It also helps determine the best locations for passenger waiting sheds and transport facilities based on real commuter activity and data.

Local government units and transport planners, the results of this study can serve as a guide in improving transport facilities and organizing jeepney routes in the area. The GIS maps and analysis can help them see where commuters gather the most and where passenger sheds or loading areas are needed. This can support better planning and decision making for public transportation in the city.

Transport cooperatives and operators, the study can help them understand commuter patterns along the Gusa to Bugo route. By knowing where passengers usually wait and travel, operators can better manage their routes, stops, and schedules to match commuter demand.

Commuters and residents, the findings of this study may help improve their daily travel experience. The recommended locations for waiting sheds and organized loading areas can provide safer and more comfortable places for passengers while waiting for jeepneys, especially during rain or strong sunlight.

Academic and professional communities, this study can serve as a reference for studies related to transportation planning, GIS mapping, and commuter behavior. It shows how spatial analysis and data collection can be used to study public transportation and improve commuter facilities in urban areas.

1.5 Conceptual Framework

This conceptual framework illustrates an integrated approach to bridging the gap between the operational values quantitative and qualitative data. By mapping independent modernization variables against intervening constraints operationally, the study designates a data-based outcome for infrastructure outcome and improved service quality.

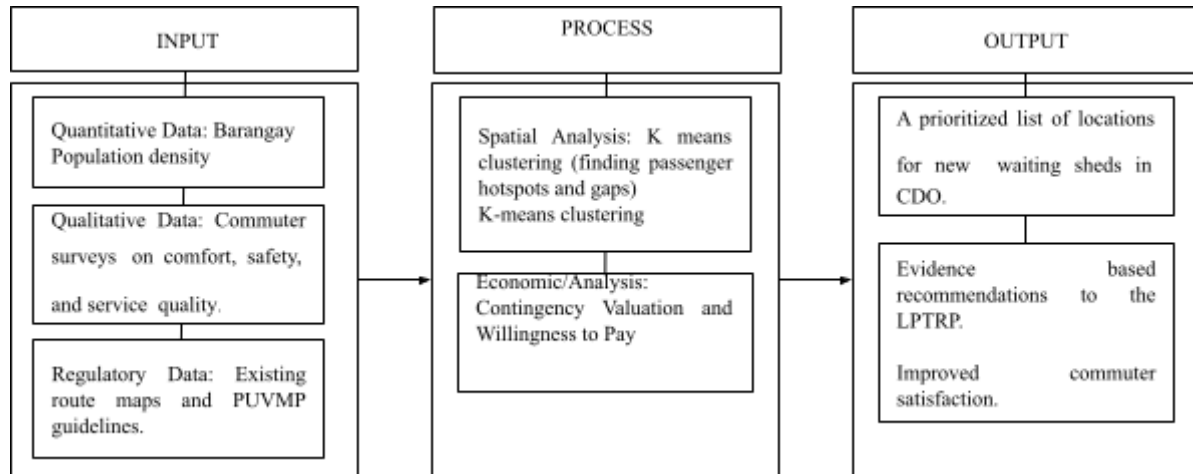


Figure 1. Conceptual Framework

1.6 Theoretical Framework

This study is grounded in transportation planning theories that emphasize the importance of data-driven infrastructure planning and spatial analysis in improving public transport systems. In urban transportation studies, the placement of passenger facilities such as stops and waiting stations should be based on commuter demand, accessibility, and spatial distribution rather than arbitrary placement. Spatial analysis is commonly used in transport planning to understand patterns of commuter movement and identify areas where passengers frequently access public transportation. Through geographic information systems (GIS) and spatial modeling techniques, transportation planners can analyze the relationship between population distribution, road networks, and commuter activity. These tools allow planners to identify locations where transportation facilities are most needed.

One important concept in transport infrastructure planning is commuter accessibility, which refers to how easily passengers can reach and use public transportation services. Accessibility is influenced by factors such as walking distance to stops, the distribution of passenger facilities, and the availability of safe and convenient waiting areas. When transportation facilities are poorly distributed, commuters may experience difficulty accessing services or may be forced to wait along unsafe roadside locations.

Another important concept is demand-based infrastructure planning, which focuses on identifying areas where passengers frequently gather or wait for transportation. By analyzing commuter movement patterns and passenger concentration points, planners can determine appropriate locations for transport waiting stations that better serve the needs of the public. In relation to this study, spatial analysis methods such as clustering techniques and geographic modeling can be used to identify commuter hotspots and service gaps in the selected barangays of Cagayan de Oro City. These methods allow researchers to determine areas where passenger demand is high and where waiting facilities are currently lacking. By

applying these concepts, the study aims to support the planning of strategically located passenger waiting stations that improve accessibility, safety, and comfort for commuters.

1.7 Scope and Limitations

This paper examines how well modernized Public Utility Jeepneys (PUJs) operate and determines where passenger waiting stations should be prioritized across six selected barangays in Cagayan de Oro City: Gusa, Cugman, Tablon, Agusan, Puerto, and Bugo. These specific barangays were selected based on their high commuter volumes, have established modernized PUJ routes running through them, and occupy important positions within the city's transportation network.

The operational evaluation concentrates on key transit performance measures such as travel time, dwell time, and passenger load factor. Researchers collect information through direct field observations and passenger surveys conducted along the selected PUJ routes that serve these barangays. This collected data helps analyze both commuter movement patterns and how vehicles perform operationally within the study areas.

Using the Geographic Information System (GIS), the research develops visual maps to identify and prioritize potential sites where passenger waiting stations may be needed. The evaluation particularly emphasizes locations with heavy passenger boarding and alighting activity, areas with efficient operations, and spots experiencing high passenger traffic.

However, this study limits itself to spatial planning considerations and does not include detailed engineering design, structural analysis, or cost calculations for the proposed waiting stations. The research also excludes examination of administrative, financial, or organizational aspects related to PUJ industry stakeholders, focusing solely on the current PUJ operational analysis. Additionally, the findings apply specifically to the selected barangays and may be influenced by various factors such as data limitations, survey response quality, traffic conditions, and changing operational conditions.

1.8 Definition of Terms

Average Travel Speed - The mean speed of a Public Utility Jeepney along a route, calculated as the total route distance divided by total travel time, including delays. It is used as an indicator of operational efficiency.

Commuter Accessibility – The measured ease of reaching and using a stop, quantified using Voronoi Diagrams to determine the effective catchment area of existing stops. Locations outside the theoretical catchment are classified as service gaps.

Commuter Demand Patterns – The spatial and temporal concentration of commuters, identified through K-means Clustering applied to the origin and destination coordinates provided in the commuter survey. The patterns also include the perceived service quality variables measured via the commuter survey.

Local Public Transport Route Plan (LPTRP) – The policy framework and guideline issued by the local government to regulate PUJ routes, stops, terminals, and supporting facilities to ensure efficient, safe, and organized public transport services.

Modernized Public Utility Jeepneys (PUJs) – A public transportation vehicle upgraded under the Philippine Public Utility Vehicle Modernization Program, equipped with improved safety features, cleaner engines, and modern technology.

Operational Efficiency Indicators - Quantitative measures (speed, dwell time, headway variation, load factor) used to evaluate PUJ service performance.

Operational Performance – The measured efficiency of PUJs using three key performance indicators (KPIs) derived from GPS telematics data: (1) Headway Variation (consistency of unit arrival spacing), (2) Average Travel Speed (travel time over distance), and (3) Dwell Time (average time spent at designated stops).

Passenger Hotspot - A location with a high concentration of boarding and alighting passengers identified through spatial clustering.

Waiting Stations – A designated space where commuters can safely wait for public transportation while being protected from weather conditions such as rain or heat.

Peak Hour - Periods of highest travel demand, typically during morning and evening commuting times.

Route Adherence - The extent to which PUJs follow their designated routes and authorized stopping points.

Service Coverage Area - The geographic area within a reasonable walking distance served by an existing PUJ stop.

Service Quality – The composite score of commuter satisfaction and perception derived from the Partial Least Squares Structural Equation Modeling (PLS-SEM) analysis. This score is based on the responses to the 24 perception questions in the Commuter Survey.

Service Gap - An area outside the effective service coverage where commuters lack adequate access to PUJ stops or facilities.

Simulation-Based Validation - The process of testing proposed facility locations using spatial and operational models to verify feasibility and effectiveness.

K-Means Clustering – A spatial analysis method that groups data points based on their similarity or proximity. In this study, it is used to identify commuter hotspots where passengers frequently gather.

Spatial Clustering - A GIS technique that groups locations based on spatial proximity to identify patterns such as demand concentrations.

Stakeholder Consultations – Meetings and discussions with relevant parties, including local government units (LGUs), transport operators, commuter representatives, and associations, to validate findings, gather input, and ensure the feasibility of proposed interventions.

Support Infrastructure – The physical facilities along the selected routes, assessed through Baseline Data Collection. This includes the mapped presence, the condition, and availability of existing waiting stations, designated loading/unloading zones, and route-specific traffic signage.

Traffic Flow Patterns – The movement dynamics of vehicles and pedestrians within the study area, including congestion points, bottlenecks, and travel delays affecting PUJ operations.

Geographic Information System (GIS) – A computer-based tool used to collect, analyze, and visualize spatial or geographic data. In this study, GIS is used to analyze commuter locations and identify suitable sites for passenger waiting stations.

Contingent Valuation Method (CVM) - A survey method used to estimate how much people value a service or improvement by asking them how much they would be willing to pay for it in a given situation.

Willingness to Pay (WTP) - The maximum amount of money that a person is willing to pay for a service, facility, or improvement based on its perceived value.

CHAPTER TWO

REVIEW OF RELATED LITERATURES AND STUDIES

2.1 Modernization and the Evolving Public Transport System

Public transportation in the Philippines has been undergoing major changes due to the implementation of the Public Utility Vehicle Modernization Program (PUVMP). This study aims to replace older jeepney units with modern vehicles that meet updated safety, environmental, and operational standards (Gatarin, 2024; Zulueta, 2024). The modernization study also seeks to improve the overall quality of transport services by introducing better route management, improved passenger comfort, and more efficient vehicle operations (Zulueta, 2024).

According to Gatarin (2024), the success of the modernization program does not depend only on replacing old jeepneys with new vehicles. It also requires proper infrastructure, route rationalization, and coordination among stakeholders such as transport operators, government agencies, and commuters. Without these supporting elements, modernization may not achieve its intended improvements in service quality and transport efficiency (Gatarin, 2024).

Studies have also shown that infrastructure limitations remain a major challenge in many Philippine cities. Many commuters still wait along roadsides without proper passenger facilities, which exposes them to heat and rain while also contributing to traffic congestion when vehicles stop randomly to load passengers (Macalalad et al., 2021; Ranosa et al., 2017). These issues demonstrate that improving both transport vehicles and commuter infrastructure is essential for creating a more reliable and organized public transportation system (Zulueta, 2024; Gatarin, 2024).

2.2 Public Transport Infrastructure and Passenger Facilities

Passenger infrastructure plays an important role in improving the efficiency and comfort of public transportation systems. Facilities such as waiting sheds, designated loading zones, and organized stops help manage commuter flow and reduce traffic disruptions (Macalalad et al., 2021; Santos et al., 2020). When these facilities are properly designed and strategically located, they help improve both commuter experience and transport efficiency (Santos et al., 2020).

According to Macalalad et al. (2021), the absence of proper waiting areas can negatively affect commuter satisfaction. Passengers who are forced to wait without shade or shelter often experience discomfort due to extreme heat, rain, or other weather conditions. This situation can discourage commuters from using public transportation and may reduce the effectiveness of modernization programs (Macalalad et al., 2021).

Ranosa et al. (2017) also highlighted that the concentration of passenger loading and unloading in certain areas can cause operational delays and traffic congestion. Their findings emphasize that proper infrastructure planning, including well-placed passenger facilities, is necessary to improve service reliability and commuter convenience (Ranosa et al., 2017).

2.3 Optimization of Bus Stop Locations Using Spatial Analysis and K-Means Clustering

The placement of bus stops and waiting sheds should be based on commuter demand and travel patterns. A study on bus stop optimization used spatial analysis combined with the K-means clustering method to determine areas with high passenger activity. This approach groups demand nodes based on their spatial distribution, allowing planners to identify optimal stop locations (Optimization Study, n.d.).

By identifying natural clusters of commuter activity, transport planners can reduce redundant stops while ensuring that passengers remain well served by the transport network. This approach also helps determine where infrastructure such as waiting sheds and loading areas will have the greatest operational impact (Optimization Study, n.d.).

Applying clustering methods to the modern jeepney system can help identify high-demand commuter areas, especially during peak hours. These insights allow planners to strategically place waiting sheds in locations where passengers frequently gather, thereby reducing walking distances and preventing overcrowding at existing stops (Optimization Study, n.d.; Santos et al., 2020).

2.4 Spatial and Environmental Analysis for Transport Waiting Shed Placement

The placement of transport shelters must also consider environmental and spatial factors. Santos et al. (2020) developed a framework for evaluating the optimal placement of bus shelters using geospatial analysis. Their study examined bus stop locations and used digital surface models to analyze shading, sunlight exposure, and surrounding urban structures (Santos et al., 2020).

The study demonstrated that environmental conditions can significantly influence the effectiveness of passenger shelters. Stops that lack proper shade or protection may expose commuters to extreme weather conditions, reducing the comfort of waiting passengers (Santos et al., 2020).

By combining spatial distribution data, environmental analysis, and commuter activity patterns, transport planners can identify locations where shelters will provide the greatest benefit. This type of data-driven planning approach ensures that commuter facilities are placed in areas where they are most needed (Santos et al., 2020; Macalalad et al., 2021).

2.5 Crowdsourced Data and Monitoring in Public Transport Systems

Modern transportation planning increasingly uses crowdsourced data to monitor transport operations and commuter behavior. According to Delgra et al. (2019), crowdsourced information collected from commuters and transport operators can provide real-time insights into transport system performance.

This type of data allows transport agencies to identify operational issues such as service delays, passenger congestion, and high-demand stops more quickly than traditional monitoring methods (Delgra et al., 2019). The use of crowdsourced transport data also improves transparency and encourages collaboration between commuters, operators, and government agencies (Delgra et al., 2019).

Integrating crowdsourced information into transport planning can help identify areas where commuters frequently wait for vehicles. These insights can support the strategic placement of

waiting sheds and other passenger facilities, ensuring that infrastructure improvements respond to actual commuter needs (Delgra et al., 2019; Santos et al., 2020).

2.6 Drivers' Perception of Jeepney Modernization

The success of the jeepney modernization study also depends on the acceptance of drivers and transport operators. Savilla et al. (2024) examined the perceptions of jeepney drivers regarding the modernization program in Indang, Cavite. Their study evaluated drivers' views on financial requirements, employment opportunities, and vehicle improvements.

The results showed that many drivers expressed concerns about the financial burden associated with modernization, particularly regarding vehicle costs, loan payments, and income stability (Savilla et al., 2024). These financial concerns contribute to resistance among some drivers toward the modernization program (Savilla et al., 2024; Gatarin, 2024).

However, the study also found that some drivers acknowledged the benefits of modernization, such as improved vehicle safety, passenger comfort, and technological features like GPS tracking and automated fare collection systems (Savilla et al., 2024). Despite these advantages, financial and livelihood concerns remain significant barriers to the full acceptance of modernization efforts (Savilla et al., 2024; Gatarin, 2024).

2.7 Environmental and Comfort Benefits of Transport Waiting Sheds

Public transport modernization also has important environmental and health benefits. Macalalad et al. (2021) emphasized that transitioning to modern public transport systems can reduce harmful emissions and improve air quality. These environmental improvements can contribute to better public health and a more sustainable urban transport system (Macalalad et al., 2021).

However, the study also highlighted that infrastructure improvements must accompany vehicle modernization. Passenger waiting areas such as bus stops and jeepney sheds are necessary to ensure commuter comfort and accessibility (Macalalad et al., 2021).

Proper waiting sheds provide protection from heat, rain, and other weather conditions. In tropical countries like the Philippines, this protection is particularly important because commuters often experience long waiting times under extreme weather conditions (Macalalad et al., 2021; Santos et al., 2020).

2.8 Climate-Responsive Design of Transport Waiting Shed

Recent research has explored innovative designs for public transport shelters that improve thermal comfort. Montero-Gutiérrez et al. (2023) developed a climate-responsive bus stop shelter that integrates radiant cooling systems powered by renewable energy sources.

Their experimental results showed that these systems significantly reduced surface temperatures and improved passenger comfort, even under high ambient temperatures (Montero-Gutiérrez et al., 2023). The cooling system reduced perceived thermal discomfort among passengers waiting at transport stops (Montero-Gutiérrez et al., 2023).

These findings suggest that modern transport waiting sheds can be designed to improve not only protection from weather but also overall commuter comfort. Integrating

climate-responsive design features such as shading, ventilation, and renewable energy technologies can significantly improve the usability of waiting sheds in hot urban environments (Montero-Gutiérrez et al., 2023; Macalalad et al., 2021).

2.9 Infrastructure and Commuter Experience

Infrastructure quality strongly influences commuter satisfaction and public transport usage. Zulueta (2024) found that commuter satisfaction with modern public utility jeepneys is strongly related to service reliability, safety, and passenger comfort.

However, the study also identified several operational inefficiencies that discourage commuters from consistently using modern jeepneys, including inconsistent route assignments and limited service coverage (Zulueta, 2024). These issues highlight the need for better transport planning and infrastructure support.

Similarly, Gatarin (2024) emphasized that modernization programs often face implementation challenges due to inadequate infrastructure, lack of coordination among stakeholders, and incomplete route rationalization. Without proper passenger facilities such as organized stops and waiting sheds, even modern vehicles may fail to provide efficient services (Gatarin, 2024).

These findings suggest that improving commuter infrastructure is essential for the success of modernization programs. Properly planned waiting sheds, organized transport stops, and well-designed passenger facilities can enhance commuter experience and support the overall efficiency of the modern jeepney system (Zulueta, 2024; Gatarin, 2024; Macalalad et al., 2021).

2.10 Contingent Valuation Method and Willingness to Pay

The Contingent Valuation Method (CVM) is a survey method used to estimate how much people are willing to pay (WTP) for improvements in public services or facilities. It is often used when the service being studied does not have a clear market price. By asking people questions about how much they would pay for certain improvements, researchers can understand how valuable the service is to them.

In transportation studies, CVM has been used to measure how much commuters are willing to pay for better transportation services. Saptutyningsih and Karimah (2019) used the contingent valuation method to estimate the willingness of passengers to pay for improvements in public transportation. Their study found that factors such as comfort, service quality, travel time, and passengers' income affect how much commuters are willing to pay. This means that when transportation services become more comfortable and reliable, commuters may be more willing to pay for the improvements.

CVM has also been used in studies in the Philippines. Fujii et al. (2016) studied the willingness of households in Metro Manila to pay for improved sewerage and sanitation services. The results showed that people's knowledge, awareness, and economic condition can influence their willingness to pay for better public services. This study shows that understanding people's opinions and financial capacity is important when planning improvements in public infrastructure.

CHAPTER THREE METHODOLOGY

3.1 Research Design

This study adopts a descriptive-analytical research design integrating primary data and secondary data supported by geospatial analysis to evaluate modern PUJ operations and identify priority locations for passenger waiting sheds in selected barangays of Cagayan de Oro City.

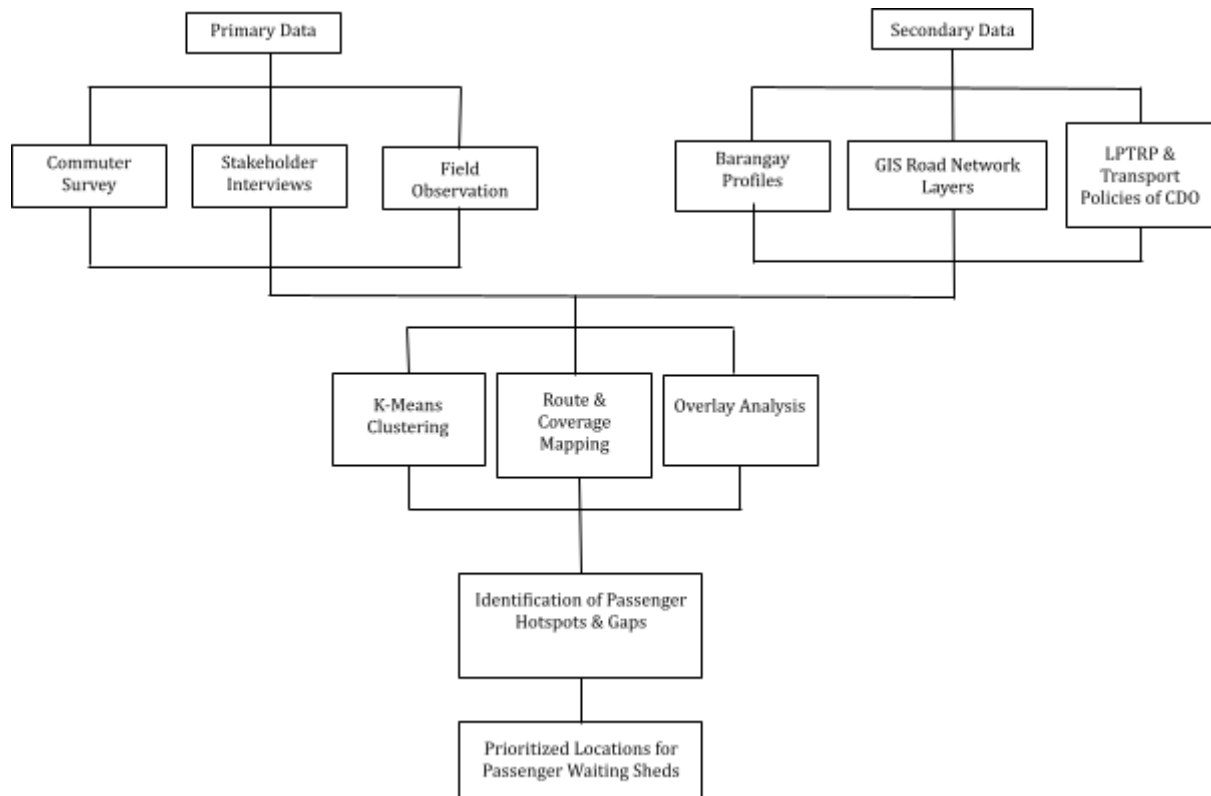


Figure 2. Research Procedure

3.2 Research Setting

This study is centered on six barangays in Cagayan de Oro City, selected based on their large population, major PUJ routes, high commuter volumes, varied land areas and road conditions. These barangays also exhibit active PUJ corridors that are essential for assessing operational performance and commuter accessibility. The chosen barangays are Gusa, Cugman, Tablon, Agusan, Puerto, and Bugo.

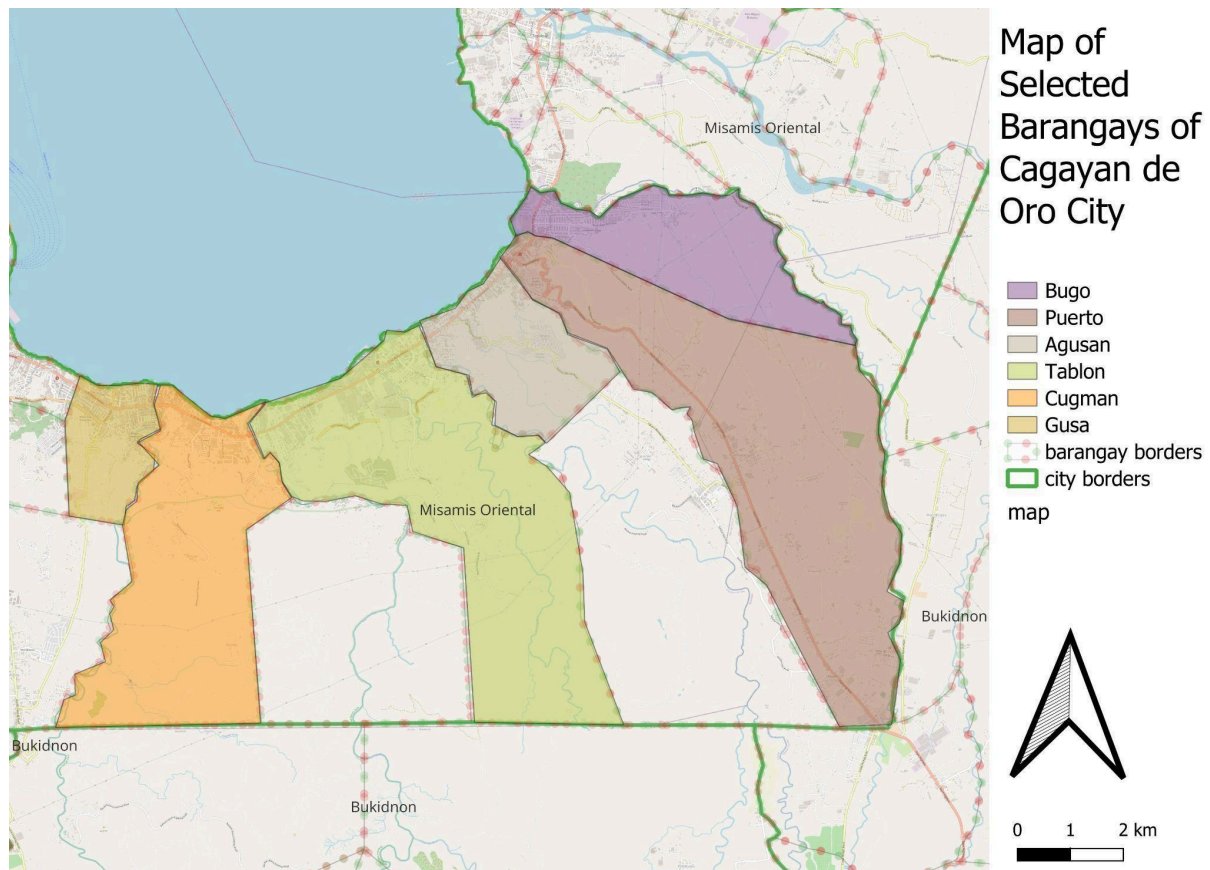


Figure 3. Research Setting

3.3 Sampling Design

The study adopts a non-probability approach incorporating spatial stratification and convenience sampling techniques to ensure representation of different commuter groups while maintaining feasibility.

The spatial sample is categorized by a 250-meter buffer zone along the National Highway, which serves as the primary corridor of PUJ operations. Within this buffer, respondents are selected from areas with high commuter activity, including roadside loading and unloading zones, commercial centers, public markets, and locations near schools and workplaces.

In order to get a representative sample of commuters, survey questionnaires will be given out across the chosen barangays; Gusa, Cugman, Tablon, Agusan, Puerto, and Bugo. The sample group will be chosen at varying times of the day, i.e. morning, midday and afternoon, such that variations in the amount of passenger demand and commuting patterns would be

captured. This will enable the research to get a wide variation of responses and mark down the frequent boarding/alighting points that will be used further to have a spatial analysis.

The sample size was determined based on feasibility, time constraints, and the need to obtain sufficient representation of commuters across the identified barangays. A minimum of 100 respondents will be targeted, with respondents proportionally distributed across the selected locations and time periods to ensure balanced data collection. Only respondents who are 18 years old and above will be included in the study.

The inclusion criteria for this study require that respondents be commuters who regularly use public utility jeepneys within the identified study area. Conversely, individuals who do not use jeepney transportation or are unwilling to participate are excluded from the study.

The information gathered will involve the normal boarding and alighting points of the respondents, their waiting period in jeepneys and participation in enhancing passenger facilities. These reactions will comprise the major data set in the form of descriptive statistical analysis, contingent valuation test and geospatial assessment to be conducted in the research.

3.4 Data Gathering Procedure

This study employs a combination of descriptive, spatial, and integrative analysis to evaluate modern PUJ service conditions and to identify priority locations for passenger jeepney terminals, loading and unloading areas, and waiting stations in the selected barangays of Cagayan de Oro City. The analysis integrates both primary and secondary data to ensure consistency with commuter experience and local transport planning policies.

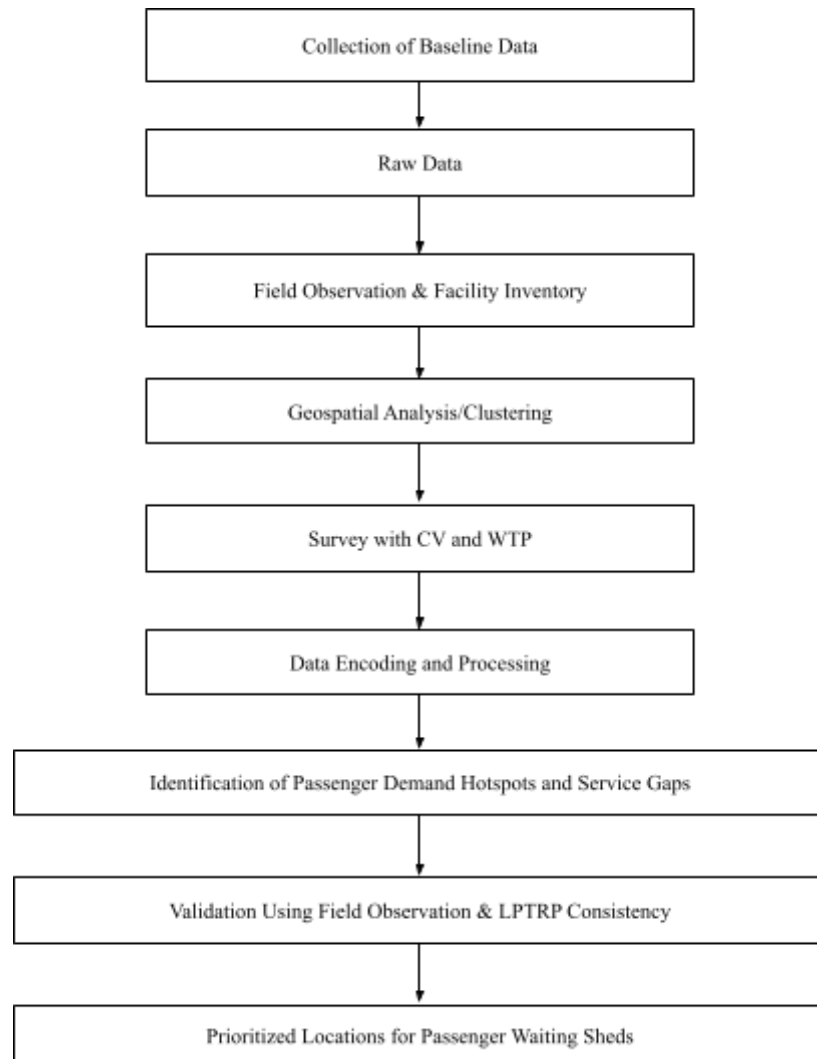


Figure 4. Data Gathering Procedure

The study will begin by collecting baseline data from the chosen 6 barangays; Gusa, Cugman, Tablon, Agusan, Puerto, and Bugo. Data obtained from the Local Public Transport Route Plan (LPTRP), barangay profiles, and route network maps are analyzed using descriptive and spatial methods. Authorized PUJ routes, terminals, and designated stopping areas are digitized and mapped using GIS to establish the official service coverage within the study area. Barangay-level data on population and land use are used to contextualize commuter demand and identify areas with high activity potential.

Field observation data are analyzed qualitatively and spatially to assess the presence, condition, and adequacy of existing waiting stations and informal boarding locations. Observed safety and accessibility issues are documented to support spatial interpretation of commuter demand patterns. Moreso, proposed identifications of spots for loading and unloading locations, waiting areas, and terminals will be identified.

Geospatial analysis is conducted using GIS software to integrate primary and secondary datasets into a unified spatial database. A 250-meter buffer around these coordinates will be applied to facility gaps. Boarding and alighting locations identified from commuter surveys are georeferenced and subjected to K-means clustering to identify passenger demand hotspots. This method groups areas with high commuter concentration and highlights locations where passenger facilities are most needed.

Authorized PUJ routes and existing passenger facilities are overlaid with the clustered demand points to assess spatial coverage and identify service gaps. Areas where high passenger demand coincides with the absence or inadequacy of waiting facilities are classified as priority zones.

To validate the data, commuter surveys are coded and summarized using descriptive statistics to determine prevailing patterns in boarding and alighting behavior, waiting conditions, perceived safety, comfort, and accessibility. Willingness to Pay (WTP) responses are analyzed using frequency distributions and central tendency measures to assess commuter acceptance of improved passenger waiting facilities.

The survey questionnaire will be prepared using both English and Bisaya languages to ensure clarity and ease of understanding among respondents in the study area. English will be used as the primary language of the instrument, while Bisaya translations will be provided for each question to accommodate local commuters who are more comfortable with the regional language.

Priority locations for commuter facilities are determined through overlay analysis by considering the following criteria: high concentration of commuter demand based on clustering results, absence or poor condition of existing waiting facilities, and proximity to authorized PUJ routes under the Local Public Transport Route Plan (LPTRP).

In addition to identifying priority locations, the study establishes a classification framework to determine the appropriate type of facility to be installed at each site, whether a waiting station, loading and unloading area, or terminal. This classification is based on the level of passenger demand, spatial characteristics, and operational function of the location. Areas identified with high passenger concentration and prolonged waiting time but without organized stopping behavior are classified as priority locations for waiting stations. Locations where frequent boarding and alighting activities cause roadside congestion and irregular stopping patterns are designated as loading and unloading zones. Meanwhile, areas that serve as major origin or destination points of routes, where multiple units start, end, or dwell for extended periods, are classified as terminals. This decision framework ensures that each proposed facility type is appropriate to the observed commuter behavior and transport operations, thereby improving both efficiency and commuter experience.

This approach ensures that recommended locations are demand-responsive, spatially justified, and aligned with local transport policies.

The identified priority locations are validated through field observations and consistency checks with LPTRP provisions to ensure practical feasibility and compliance with existing route plans. The results from descriptive analysis, spatial clustering, and policy alignment are synthesized to support evidence-based recommendations for passenger waiting station placement and improvements in PUJ service accessibility.

3.5 Research Instruments and Standards

The research instruments to be used in this study in collecting data include three main instruments namely survey questionnaires, field observation sheets, and secondary data records.

The main tools utilized in the research is a structured commuter survey questionnaire that will be used to communicate data on passenger travel behavior, waiting experience, and views on the current jeepneys services. The survey will have four items, including profile of the respondents, commuting, quality of service perception, and readiness to spend money on better passenger waiting areas. These perception questions identify the variables of the perception of comfort, safety, accessibility, reliability, and waiting experience on a five-point Likert scale, which would have a strongly disagree to strongly agree scale.

The survey also considers Contingent Valuation (CV) and Willingness-To-Pay (WTP) questions to determine the commuter acceptance of the improved facilities and the respondents are asked to give the amount they can give in case of improved passenger waiting stations. This is a standard approach to evaluating the value of service enhancements by users in transportation research.

Field observation checklists will also be used in order to carry out systematic recording of the physical conditions and availability of passenger waiting stations, loading and unloading points, pedestrian accessibility, and safety conditions at target locations. The investigators will carry out observations as part of site visits in each barangay.

To provide some contextual information about the current PUJ routes and infrastructure in the area of study, secondary data will be collected by local government repositories, such as barangay profiles, route maps, and LPTRP.

All the study tools will be in line with established ethical principles in academic research. Respondents are free to take part in the survey, and no personal identifying information disclosure is obligatory. The research will adhere to the stipulations of the Data Privacy Act of 2012 to maintain confidentiality and handling of data in a responsible manner.

Participants will not receive any monetary or material compensation for their participation in the study. Their involvement is entirely voluntary.

The researchers will provide an informed consent form to all participants before the survey is conducted. The form will explain the purpose of the study, procedures involved, voluntary nature of participation, and assurance of confidentiality. Participants will be given sufficient time to read and understand the information before agreeing to take part in the study. The study involves minimal risk to participants as it only requires them to answer a survey

questionnaire about their commuting experience. Possible risks may include slight inconvenience, time consumption, or minor discomfort in answering some questions.

Participants will have the right to withdraw from the study at any point without any penalty or consequence. If a participant chooses to withdraw, any data provided by the participant will be excluded from the analysis upon request. The researchers will respect the decision of the participants and ensure that their rights are protected throughout the study. The participants have the right to access their own responses and records related to the study upon request, subject to reasonable conditions and data privacy regulations.

3.6 Mode of Data Analysis

The data of the study will be analyzed using descriptive statistics, contingent valuation, and GIS. These procedures will evaluate commuter behavior, plot passenger demand and identify priority locations of waiting stations in the chosen barangays in Cagayan de Oro City.

The summaries of the commuter survey will be first described using descriptive statistics. The outcomes will be presented in the form of frequency tables, percentages, and averages that will give a general picture of commuting conditions. Some of the variables that we will look at include waiting time, boarding sites, and travel behavior in order to present a clear picture of the commuting experience. The collected data will be used strictly for academic purposes and will not be shared with third parties without permission.

In order to estimate the importance, the commuters attach to the improved facilities, we will use the Contingent Valuation Method (CVM). This approximates readiness to pay the enhancement such as waiting stations. We shall interpret the survey data using descriptive statistics and identify the mean willingness to pay and support of the proposed upgrades.

Passenger activity will be mapped with the help of GIS software, including QGIS or ArcGIS. The surveys will be converted into boarding and alighting spots, which will be used to create the coordinates and plotted on a digital map of the study area. This mapping allows scholars to observe cartographic relations, as well as identify where the facilities are in most demand.

K-means clustering of the georeferenced information will then be used. These clusters point together by distance and relatedness demonstrating an agglomeration of demand. The algorithm will determine three clusters of high, moderate, and low demand which will indicate hotspots that must be given priority when new waiting stations are being set up.

Lastly, we will integrate clustering findings and survey data to make data-driven decisions. The study will be used to locate strategic points to position passenger waiting stations in order to improve safety, accessibility and general efficiency of the transport modes within the selected barangays by incorporation of the demand patterns, willingness to pay information and spatial clusters.

3.7 Budget Plan

Following is Table 1 for the estimated budget plan for the duration of the study;

Table 1. Budget Plan

I.	Fieldwork and Data Collection	
	A. Transportation (Land Travel)	₱ 3,000.00
	B. Survey Questionnaires	₱ 3,000.00
II.	Research Materials	
	A. Final paper	₱ 1,500.00
	B. Printer Ink	₱ 2,000.00
	C. Bondpaper	₱ 500.00
TOTAL:		₱ 10,000.00

3.8 Plan of Implementation

Figure 5 indicates the intended schedule of the research. The schedule starts in December 2025 and ends in November 2026. The research will launch in December 2025. During this month, the research proposal will be prepared, and questionnaires on survey questions will be formulated. A budget of 10,000 Philippine peso will be allocated for this study. From the initial planning phase to survey printing/administration costs, and preliminary travel expenses for field assessment. The survey questionnaires will be worked on until January 2026, confirming coverage of all research variables. The survey questionnaires will be handed over and collected between June and July 2026, whereas a key informant interview will be carried out to ensure that the stakeholders' feedback is obtained.

To support and corroborate the previous data, further data collection will be implemented in July 2026. All the acquired information will be computed and analyzed within the timeframes of August to September 2026 to obtain meaningful outcomes. Lastly, the Final Report will be drafted between October and November 2026, and the compilation of all findings, analyses, and recommendations will take place, to be submitted and presented.

	2025	2026										
ACTIVITY	DEC	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV
Research Proposal												
Survey Questionnaires Formulation												
Draft of Questionnaire and IREB												
Research Ethics Office Submission: Review and Revision												
Baseline Data Collection												
Data Collection: GPS Telematics												
Deployment of Questionnaire for Commuter												
Distribution Of Questionnaires for Operators and drivers												
Collection Of Survey Questionnaires												
Data Collection												
Computation/ Data Analysis												
Drafting Of Final Report												

Figure 5. Proposed Project Timeline

CHAPTER FOUR EXPECTED RESULTS AND DISCUSSION

4.1 Socio-Demographic Profile of the Respondents

This section presents the demographic characteristics of the respondents who participated in the survey. Understanding the age and occupation of commuters provides important background information that helps explain travel behavior and transportation preferences. These variables also establish whether the survey respondents adequately represent the major users of modern jeepney transportation along the study corridor.

4.1.1 Age Distribution of Respondents

Table 2. Age Distribution of Respondents

Age Group	Frequency	Percentage (%)
Below 18	22	5.71
18–25	186	48.31
26–35	84	21.82
36–45	56	14.55
46 years old and above	37	9.61
Total	385	100.00

Table 4.1 presents the age distribution of the 385 respondents who regularly commute using modern jeepneys along the Barangay Gusa to Barangay Bugo corridor. The largest proportion of respondents belonged to the 18–25 years old age group, comprising 186 respondents or 48.31% of the total sample. This indicates that nearly one-half of the commuters are young adults who regularly depend on public transportation for educational, employment, and personal travel purposes.

The 26–35 years old age group accounted for 84 respondents (21.82%), followed by commuters aged 36–45 years old with 56 respondents (14.55%). Meanwhile, respondents aged 46 years old and above represented 9.61% of the total respondents, while only 5.71% were below eighteen years old.

The results suggest that modern jeepneys primarily serve economically active individuals and students, which is consistent with the presence of educational institutions, commercial establishments, and employment centers along the Gusa–Bugo transportation corridor. Consequently, improvements to commuter facilities are expected to benefit the largest segment of regular public transport users.

4.1.2 Occupation of Respondents

Table 4.2 Occupation of Respondents (n = 385)

Occupation	Frequency	Percentage (%)
Student	165	42.86
Private Employee	109	28.31
Government Employee	33	8.57
Self-employed / Business Owner	46	11.95
Homemaker / Unemployed	21	5.45
Others	11	2.86
Total	385	100.00

Table 4.2 presents the occupational profile of the respondents. Students comprised the largest group with 165 respondents (42.86%), followed by private sector employees with 109 respondents (28.31%). Self-employed individuals accounted for 46 respondents (11.95%), while government employees represented 33 respondents (8.57%).

Only 21 respondents (5.45%) identified themselves as homemakers or unemployed, whereas 11 respondents (2.86%) belonged to other occupational categories.

The results indicate that students and employed individuals constitute the primary users of modern jeepney transportation within the study area. Their daily dependence on public transportation highlights the importance of reliable transport services, efficient travel times, and adequate waiting facilities. Since these commuters travel regularly, improvements in transportation infrastructure have the potential to significantly enhance accessibility, safety, and commuting convenience.

4.2 Existing Modern Jeepney Routes and Travel Characteristics

This section presents the travel characteristics of respondents, including the modern jeepney routes utilized, boarding locations, and destination points. These data establish the spatial travel patterns of commuters and provide the basis for subsequent Geographic Information System (GIS) analysis and K-means clustering.

4.2.1 Modern Jeepney Routes Utilized

Table 4.3 Modern Jeepney Routes Frequently Used by Respondents

Route	Frequency	Percentage (%)
Bugo–Cogon	137	35.58
Gusa–Cogon	112	29.09
Puerto–Cogon	78	20.26
Other Modern Jeepney Routes	58	15.06
Total	385	100.00

Table 4.3 summarizes the modern jeepney routes most frequently utilized by the respondents. The Bugo–Cogon route recorded the highest number of commuters with 137 respondents (35.58%), followed by the Gusa–Cogon route with 112 respondents (29.09%). Meanwhile, Puerto–Cogon accounted for 20.26%, while the remaining respondents used other authorized modern jeepney routes.

The findings indicate that commuter demand is concentrated along routes connecting eastern residential barangays to the central business district of Cagayan de Oro City. These routes experience continuous passenger movement throughout the day and therefore require efficient passenger loading facilities and improved waiting areas to accommodate commuter demand.

4.2.2 Boarding Locations of Respondents

Table 4.4 Common Boarding Locations

Barangay	Frequency	Percentage (%)
Bugo	83	21.56
Puerto	73	18.96
Gusa	70	18.18
Tablon	62	16.10
Cugman	54	14.03
Agusan	43	11.17
Total	385	100.00

The spatial distribution of boarding locations reveals that Bugo recorded the highest number of boarding passengers, accounting for 21.56% of the respondents. Puerto and Gusa followed

with 18.96% and 18.18%, respectively. These findings suggest that the eastern section of the study corridor serves as a major origin point for commuters traveling toward commercial and institutional areas.

The concentration of boarding activity in these barangays indicates increased demand for organized passenger waiting facilities. Existing roadside waiting practices observed in these locations may contribute to traffic interruptions and reduced commuter safety, emphasizing the need for designated waiting sheds and loading areas.

4.2.3 Destination (Alighting) Locations

Table 4.5 Common Destination Locations of Respondents

Destination	Frequency	Percentage (%)
Cogon Commercial District	134	34.81
Puerto	69	17.92
Bugo	42	10.91
Agusan	40	10.39
Tablon	37	9.61
Cugman	35	9.09
Gusa	28	7.27
Total	385	100.00

As shown in Table 4.5, the Cogon Commercial District emerged as the most common destination, representing 34.81% of all respondents. This finding confirms that the majority of commuters travel toward the city's principal commercial and employment center. The remaining destinations were distributed among the six barangays included in the study, reflecting both inbound and outbound travel patterns.

The spatial distribution of destination points demonstrates that commuter movements are highly concentrated along the National Highway corridor, particularly toward areas characterized by commercial establishments, educational institutions, and government offices. These travel patterns provide an important foundation for the GIS-based hotspot analysis and K-means clustering presented in the succeeding sections of this chapter.

4.3 Commuter Perception on Fare, Distance, Delay, and Service Satisfaction

This section presents the respondents' perceptions regarding the affordability of modern jeepney fares, estimated travel distance, travel delays experienced during peak hours, and their overall level of satisfaction with selected service attributes. These findings address the second objective of the study, which seeks to evaluate commuter perceptions of the current modern jeepney transportation system.

4.3.1 Modern Jeepney Fare

Table 4.6 Distribution of Respondents According to Fare Paid (n = 385)

Fare Category	Frequency	Percentage (%)
Minimum Fare (₱15.00)	132	34.29
₱16.00–₱25.00	168	43.64
₱26.00–₱35.00	64	16.62
More than ₱35.00	21	5.45
Total	385	100.00

Table 4.6 presents the fare distribution of the respondents. The majority of commuters (43.64%) reported paying between ₱16.00 and ₱25.00 per trip. Meanwhile, 34.29% paid only the minimum fare of ₱15.00, while fewer respondents paid higher fares due to longer travel distances.

The results indicate that most commuters travel moderate distances within the study corridor. Despite recent fare adjustments under the Public Utility Vehicle Modernization Program, the fare structure remains generally acceptable to commuters, particularly when compared to the convenience and comfort offered by modern jeepneys.

4.3.2 Estimated Travel Distance

Table 4.7 Estimated Travel Distance per Trip

Distance Category	Frequency	Percentage (%)
Less than 4 km	82	21.30
4 km–8 km	185	48.05
More than 8 km	118	30.65
Total	385	100.00

As shown in Table 4.7, nearly half of the respondents (48.05%) traveled between 4 and 8 kilometers during a single trip. Approximately 30.65% traveled more than 8 kilometers, while only 21.30% traveled less than 4 kilometers.

These findings demonstrate that the majority of commuters depend on modern jeepneys for medium- to long-distance travel. Consequently, passenger comfort, efficient travel time, and the availability of adequate waiting facilities become increasingly important, particularly for commuters who spend considerable time traveling each day.

4.3.3 Travel Delay Experienced During Peak Hours

Table 4.8 Average Delay Experienced During Peak Hours

Delay Category	Frequency	Percentage (%)
Minimal Delay (0–5 minutes)	48	12.47
Moderate Delay (6–15 minutes)	163	42.34
Heavy Delay (16–30 minutes)	126	32.73
Severe Delay (More than 30 minutes)	48	12.47
Total	385	100.00

The results indicate that 42.34% of respondents experienced delays ranging from 6 to 15 minutes, while 32.73% experienced delays between 16 and 30 minutes during peak travel periods.

Only 12.47% reported minimal delays, whereas another 12.47% experienced delays exceeding thirty minutes. These findings suggest that traffic congestion and passenger loading activities remain significant factors affecting travel efficiency along the Gusa–Bugo corridor.

The prevalence of moderate and heavy delays emphasizes the need for properly designated loading zones and organized passenger waiting areas. Such facilities can reduce roadside congestion and improve traffic flow by minimizing random stopping and passenger accumulation along the highway.

4.3.4 Level of Satisfaction with Modern Jeepney Service Attributes

To determine commuter satisfaction with the current modern jeepney service, respondents rated three service attributes using a five-point Likert scale, where 5 indicates Very Satisfied and 1 indicates Very Dissatisfied.

The weighted mean for each service attribute was computed using the following formula:

$$\bar{x} = \frac{\sum fx}{N}$$

where:

$\bar{x}$ = Weighted Mean

f = Frequency

x = Rating Scale

N = Total Number of Respondents

The following verbal interpretation was adopted:

Weighted Mean	Interpretation
4.21–5.00	Very Satisfied
3.41–4.20	Satisfied
2.61–3.40	Moderately Satisfied
1.81–2.60	Dissatisfied
1.00–1.80	Very Dissatisfied

Table 4.9 Level of Satisfaction with Modern Jeepney Service

Service Attribute	Weighted Mean	Verbal Interpretation	Rank
Reasonable Fare Rate	4.14	Satisfied	1
Travel Time and Minimal Delays	3.38	Moderately Satisfied	2
Safety and Availability of Waiting Sheds	2.61	Moderately Satisfied	3
Overall Mean	3.38	Moderately Satisfied	

Table 4.9 presents the respondents' level of satisfaction regarding selected attributes of the modern jeepney transportation system. Among the evaluated indicators, Reasonable Fare Rate obtained the highest weighted mean of 4.14, corresponding to a verbal interpretation of Satisfied. This result indicates that commuters generally perceive the current fare structure as fair and affordable relative to the quality of service provided.

Travel Time and Minimal Delays obtained a weighted mean of 3.38, interpreted as Moderately Satisfied. Although respondents acknowledged the improvements brought about by modern jeepneys, recurring traffic congestion and prolonged passenger loading activities continue to affect overall travel efficiency.

The Safety and Availability of Waiting Sheds recorded the lowest weighted mean of 2.61, which falls under the Moderately Satisfied category but is very close to the threshold for dissatisfaction. This suggests that while some commuter facilities are available, respondents perceive them to be insufficient in terms of quantity, accessibility, maintenance, and protection from adverse weather conditions.

Overall, the service attributes yielded an average weighted mean of 3.38, indicating that respondents were Moderately Satisfied with the present modern jeepney transportation system. The findings imply that while fare affordability is viewed positively, improvements in passenger infrastructure and traffic management remain necessary to enhance the overall commuting experience.

4.4 Contingent Valuation Method (CVM) and Willingness to Pay

This section presents the respondents' willingness to contribute a nominal maintenance fee for the continuous operation, maintenance, and improvement of modern passenger waiting sheds, designated loading zones, and transport terminals. The Contingent Valuation Method (CVM) was employed to estimate the monetary value commuters place on improved transportation facilities.

4.4.1 Willingness to Contribute a Maintenance Fee

Table 4.10 Respondents' Willingness to Pay for Improved Commuter Facilities

Response	Frequency	Percentage (%)
Yes	316	82.08
No	69	17.92
Total	385	100.00

Table 4.10 shows that 316 respondents (82.08%) expressed their willingness to contribute a small maintenance fee for the continued improvement of commuter facilities. In contrast, 69 respondents (17.92%) indicated that they were unwilling to provide additional financial contributions.

The high percentage of positive responses demonstrates that commuters recognize the value of investing in safe, clean, and well-maintained waiting sheds and loading facilities. This finding further suggests that respondents believe improvements in commuter infrastructure will enhance travel convenience, passenger safety, and overall commuting efficiency.

4.4.2 Maximum Amount Willing to Pay

Respondents who answered "Yes" to the previous question were further asked to indicate the maximum amount they were willing to contribute per trip for the continuous maintenance and operation of commuter facilities.

Table 4.11 Maximum Amount Respondents are Willing to Contribute (n = 316)

Amount	Frequency	Percentage (%)
₱2.00	142	44.94
₱5.00	121	38.29
₱10.00	39	12.34
More than ₱10.00	14	4.43
Total	316	100.00

Table 4.11 shows that the largest proportion of respondents (44.94%) were willing to contribute ₱2.00 per trip for the maintenance of commuter facilities. This was followed by 38.29% who selected ₱5.00 as their preferred contribution.

Only 12.34% expressed willingness to contribute ₱10.00, while 4.43% indicated a willingness to pay more than ₱10.00 per trip.

These findings suggest that commuters generally support transportation infrastructure improvements provided that the required contribution remains affordable. The results indicate that implementing a minimal maintenance fee may be socially acceptable if accompanied by visible improvements in commuter facilities and service quality.

4.4.3 Estimated Mean Willingness to Pay

The average willingness to pay (WTP) was computed using the weighted average formula.

$$\bar{x} = \frac{\sum fx}{N}$$

Using the survey responses,

$$\bar{x} = \frac{(142)(2) + (121)(5) + (39)(10) + (14)(15)}{316} = \text{₱}4.72$$

Therefore, the estimated mean willingness to pay is ₱4.72 per commuter per trip.

The computed average indicates that commuters place measurable economic value on improved passenger waiting facilities. This result demonstrates public support for infrastructure enhancement initiatives provided that the additional contribution remains minimal and transparent.

4.5 Geographic Information System (GIS) Spatial Analysis

Geographic Information System (GIS) analysis was conducted to determine the spatial distribution of commuter movement within the study area. The collected boarding and alighting locations were georeferenced and mapped to identify areas with high passenger activity.

The resulting spatial analysis provides a visual representation of commuter travel patterns and serves as the primary input for the K-means clustering analysis.

4.5.1 Spatial Distribution of Boarding Locations

*Figure 4.11 GIS Map of Boarding Locations
(Insert GIS Boarding Point Map)*

The GIS-generated boarding point map revealed that passenger origins were primarily concentrated in the barangays of Bugo, Puerto, and Gusa. These locations exhibited the highest density of commuter boarding activities because of their large residential populations and proximity to major road corridors.

Conversely, Agusan and Cugman recorded fewer boarding activities, indicating relatively lower passenger demand.

The spatial concentration of boarding locations demonstrates that commuter demand is unevenly distributed across the study corridor, thereby emphasizing the need for strategically located passenger waiting facilities.

4.5.2 Spatial Distribution of Destination Points

*Figure 4.12 GIS Map of Alighting Locations
(Insert GIS Destination Map)*

The GIS destination map indicates that a significant proportion of commuters travel toward Cogon Commercial District, which functions as the city's primary commercial and employment center.

Other destination points include Puerto, Bugo, Tablon, Agusan, Cugman, and Gusa.

The clustering of destination points around commercial establishments suggests that passenger demand is strongly influenced by employment opportunities, educational institutions, government offices, and business districts.

4.5.3 Passenger Density Map

*Figure 4.13 Passenger Density Heat Map
(Insert Heat Map)*

The passenger density map illustrates varying levels of commuter concentration throughout the study area. High-density passenger movements were observed along the National Highway, particularly near intersections, commercial centers, schools, and transport terminals.

These areas experience frequent passenger loading and unloading activities throughout the day, making them suitable locations for future commuter waiting sheds and designated loading areas.

4.6 K-means Clustering Analysis

K-means clustering was employed to classify commuter activity into homogeneous groups based on their boarding and destination coordinates.

Three clusters were generated to categorize passenger demand according to intensity.

Table 4.12 K-means Clustering Results

Cluster	Demand Level	Number of Passenger Points
Cluster 1	High Demand	162
Cluster 2	Moderate Demand	134
Cluster 3	Low Demand	89
Total	385	

The K-means clustering analysis successfully categorized commuter activity into three distinct demand groups.

Cluster 1 represented high-demand commuter areas, accounting for 162 passenger points. These locations exhibited frequent boarding and alighting activities and were primarily situated along major commercial corridors.

Cluster 2 consisted of 134 passenger points representing moderate commuter demand. These locations generally served residential communities with consistent but relatively lower passenger volumes.

Meanwhile, Cluster 3, containing 89 passenger points, represented low-demand areas characterized by limited commuter activity.

The clustering analysis demonstrates that passenger demand is spatially concentrated rather than uniformly distributed throughout the transportation corridor. Consequently, transportation infrastructure investments should prioritize high-demand locations to maximize commuter benefits.

4.7 Proposed Priority Locations for Modern Passenger Waiting Sheds

The proposed waiting shed locations were determined by integrating the following datasets:

- GIS passenger density analysis
- Boarding and destination maps
- K-means clustering results
- Delay analysis
- Commuter satisfaction ratings
- Willingness-to-pay analysis

Table 4.13 Proposed Priority Locations for Passenger Waiting Sheds

Proposed Location	Cluster	Priority Level	Justification
Bugo Public Market	High	Very High	Highest boarding demand

Puerto National Highway	High	Very High	Major transfer corridor
Gusa National Highway	High	High	High passenger volume
Tablon Crossing	Moderate	High	Frequent loading activities
Agusan Junction	Moderate	Moderate	Moderate commuter demand
Cugman Highway	Low	Moderate	Future service expansion

The proposed waiting shed locations were selected based on the integration of commuter demand, spatial analysis, and public perception. Areas identified as high-demand clusters demonstrated the greatest need for organized passenger facilities due to the volume of boarding and alighting activities.

Bugo Public Market and Puerto National Highway emerged as the highest-priority locations because they serve as major origins and destinations for daily commuters. These areas experience continuous passenger movement and currently exhibit inadequate waiting facilities.

The installation of modern waiting sheds at these locations is expected to improve passenger comfort, enhance commuter safety, reduce roadside congestion, and facilitate more efficient passenger loading and unloading operations.

4.8 Summary of Findings

The results of the study revealed that the majority of the respondents were young adults between 18 and 25 years old, with students comprising the largest occupational group among modern jeepney commuters. This demographic profile indicates that the modern jeepney system primarily serves individuals who rely on public transportation for education, employment, and other daily activities. The findings further showed that the Bugo–Cogon route recorded the highest commuter utilization, while the barangays of Bugo, Puerto, and Gusa served as the primary boarding locations. Most respondents identified the Cogon Commercial District as their destination, demonstrating that commuter movement is largely concentrated between residential areas and the city's commercial and employment center.

The analysis of travel characteristics revealed that most respondents traveled moderate distances of four to eight kilometers and generally paid fares ranging from ₱16.00 to ₱25.00 per trip. Moderate travel delays of six to fifteen minutes were the most frequently experienced during peak hours, suggesting that traffic congestion and passenger loading activities continue to affect travel efficiency along the study corridor. Although respondents expressed satisfaction with the current fare structure, the overall evaluation of the modern jeepney service indicated only moderate satisfaction, primarily due to concerns regarding travel delays and the limited availability of safe and convenient passenger waiting sheds.

The Contingent Valuation Method showed that a large majority of commuters were willing to contribute a nominal maintenance fee to support the installation and continuous maintenance of modern commuter facilities. Most respondents preferred a minimal contribution ranging from ₱2.00 to ₱5.00 per trip, resulting in an estimated average willingness to pay of ₱4.72. This finding suggests that commuters recognize the importance of investing in transportation

infrastructure, provided that the proposed fee remains affordable and is directly utilized for improving commuter facilities.

The Geographic Information System (GIS) analysis and K-means clustering successfully identified areas with varying levels of commuter demand throughout the study corridor. High passenger concentrations were observed in Bugo, Puerto, and Gusa, indicating that these locations experience the greatest boarding and alighting activities. Based on the integration of commuter demand, spatial analysis, satisfaction ratings, and willingness-to-pay results, these areas were identified as the highest-priority locations for the installation of modern passenger waiting sheds. Overall, the findings demonstrate that combining commuter perception, economic valuation, and spatial analytical techniques provides an effective basis for identifying priority locations for transportation infrastructure improvements and supports evidence-based planning for the modernization of commuter facilities along the Gusa–Bugo corridor in Cagayan de Oro City.

BIBLIOGRAPHY

- Abad, S. K., & Abad, K. A. (2023). Synergizing geographic information systems (GIS) and multicriteria decision making analysis (MCDA) for public transit network optimization: A review. *Scholar's Digest- Journal of Multidisciplinary Studies*, 3(2), 173–182
- Abhilash, S. S. S., Sreedevi, M., & Rao, K. R. (2019). Public transportation service quality assessment using AVL and fare card data. *Transportation Research Part A: Policy and Practice*, 125, 245–256
- Agaton, C. B., et al. (2023). State of the art: Public utility vehicle modernization program in the Philippines. *Journal of Sustainable Development in Transport & Infrastructure*, 4(2), 45-60.
- Alharbi, M., & Hassan, M. (2023). Optimizing the distribution of public transportation stops using GIS: A case of Amman City in Jordan. *Civil Engineering and Architecture*, 11(3), 1558–1568. <https://doi.org/10.13189/cea.2023.110335>
- Ali, A. B. (2018). Optimizing bus stop location and spacing using the Analytical Hierarchy Process (AHP) and GIS. *International Journal of Transportation Science and Technology*, 7(1), 1–11
- Atos, M. S. Q., Cabe, K. B., Gomez, C. J. E., & Padupad, C. L. O. (2021). Modernized tradition: Transformation of public transport. DLSU Senior High School Research Congress. Animo Repository. https://animorepository.dlsu.edu.ph/conf_shsrescon/2021/paper_spl/2/
- Bacero, R. M., Bandala, A. A., & Fillone, A. M. (2017). Proposed jeepney fare determination using fuzzy logic. *Philippine Engineering Journal*, 38(2), 23–35. <https://ncts.upd.edu.ph/tssp/wp-content/uploads/2017/07/TSSP2017-08-Bacero-Bandala-and-Fillone.pdf>
- Balingit, J. C. (2022). The missing link: Assessing the roadside infrastructure of the PUV modernization program. *Philippine Journal of Public Administration*, 66(1), 88-105.
- Bureau of Philippine Standards. (2017). Public utility vehicles – Class 2 and Class 3 – Specifications (PNS 2126:2017). Department of Trade and Industry.
- Cagayan de Oro City Council. (2022). City Ordinance No. 14337-2022: An ordinance adopting the Local Public Transport Route Plan (LPTRP) of Cagayan de Oro City. City Government of Cagayan de Oro.
- Cahigas, M. M. L., Zulvia, F. E., Ong, A. K. S., & Prasetyo, Y. T. (2023). A comprehensive analysis of clustering public utility bus passenger's behavior during the COVID-19 pandemic: Utilization of machine learning with metaheuristic algorithm. *Sustainability*, 15(9), 7410. <https://doi.org/10.3390/su15097410>
- Cueto, M. J., Baluyot, M. M., Ordonia, R. L., & Rivera, J. P. (2024). Intelligent transport system implementation in Manila: A four-step model approach. *International Journal of Transportation Science and Technology*, 13(1), 88–102. <https://ncts.upd.edu.ph/tssp/wp-content/uploads/2024/12/TSSP2024-11-Revised-Paper.pdf>
- Cunanan, A. L. (2025). Data-driven jeepney modernization: Leveraging telematics data for public transport indicators. SafeTravelPH Mobility Group.
- Delgra, M. A., et al. (2019). Public transport crowdsourcing and collaborative governance for BRT in transition: Monitoring and analyzing the EDSA Carousel in Metro Manila.

Transportation Research Procedia, 41, 268–279.

https://easts.info/on-line/proceedings/vol.13/pdf/PP3045_R1_F.pdf

Department of Transportation. (2017). Omnibus guidelines on the planning and identification of public road transportation services and franchise issuance (Department Order No. 2017-011). Government of the Philippines.

Du, J., Shi, J., Zhang, N., Liu, Y., & Liu, W. (2025). Correction: Du et al. Preliminary Establishment of an Efficient Regeneration and Genetic Transformation System for *Hemerocallis middendorffii* Trautv. & C. A. Mey. *Horticulturae*, 11(5), 528.

<https://doi.org/10.3390/horticulturae11050528>

Duan, Z., et al. (2018). Using GPS data to assess operational performance of public bus routes. *Transportation Research Part C*, 86, 374–389.

Francesquite, R. J., & Calva, J. P. (2024). The unveiling of the e-jEEPney modernization program through the lenses of diverse sectors in Southern Philippines. *Sustainability*, 16(9), 3912. <https://journal.formosapublisher.org/index.php/eajmr/article/view/8860?>

Francisco, K. (2025). Transport infrastructure in the Philippines: From plans to actual allocation. Philippine Institute for Development Studies. <https://doi.org/10.62986/dp2024.51>

Gatarin, E. R. (2024). Socio-economic and operational challenges of the Philippine PUV modernization program. *Journal of Transport Policy and Planning*, 12(2), 41–56. https://cids.up.edu.ph/wp-content/uploads/2024/08/Examining_PUVMP_Just_Transition_Lens_JEEPney_Operators_Bacolod3.pdf

Goud, S. D. (2020). The impact of infrastructure on transit service reliability: A case study. *Journal of Public Transportation*, 22(1), 1–15

Gumasing, M. J. J., et al. (2024). Factors influencing the adoption of electric jeepneys in the Philippines. *Transportation Research Interdisciplinary Perspectives*, 19, 100832. <https://www.mdpi.com/2032-6653/15/7/284>

Gumasing, M. J. J., et al. (2024). Is modernization of public transport in the Philippines a boon or a bane for the Filipino people? *Management Research and Innovation Journal*, 20(1), 4-26.

Gumasing, M. J. J., Tadina, K. P. V., & Vidamo, N. D. (2021). Sustainability model of E-JEEPney operations in Paranaque, Metro Manila. In 2021 IEEE 8th International Conference on Industrial Engineering and Applications (ICIEA) (pp. 247–251). IEEE. <https://doi.org/10.1109/ICIEA52957.2021.9436813>

Japan International Cooperation Agency. (2019). The roadmap for transport infrastructure development for Metro Manila and its surrounding areas. JICA Reports.

Kock, N., & Hadaya, P. (2018). Minimum sample size estimation in PLS-SEM: The inverse square root and gamma-exponential methods. *Information Systems Journal*, 28(1), 227–261. <https://doi.org/10.1111/isj.12131>

Labrador, C. L., Palma, A. J., Santiago, M. R., Torres, J. D., Gaviola, J. A., & Muhi, R. A. (2010). Traffic flow impact of queuing public utility jeepneys at SM City Sta. Mesa. *Journal of the Eastern Asia Society for Transportation Studies*, 8, 1435–1449. https://ncts.upd.edu.ph/tssp/wp-content/uploads/2017/02/2010-08_labrador_final.pdf

Lalio, J. A., & Navarro, R. S. (2025). Commuters' preferences and willingness to pay for modern public utility vehicles. *Asian Transport Studies*, 9, 100134. https://www.researchgate.net/publication/390126481_COMMUTERS%27_PREFERENCES

AND THEIR WILLINGNESS TO PAY FOR MODERN PUBLIC UTILITY VEHICLE MPUV

Li, Y., Wang, X., & Zhang, H. (2023). Study on location of bus stops in subway service areas based on residents' travel accessibility. *Sustainability*, 15(6), 4873.

<https://www.mdpi.com/2071-1050/15/5/4517?>

Macalalad, A. A., Tolentino, N. J., et al. (2021). Environmental and socio-economic benefits of public transport modernization in the Philippines. *Transportation Research Interdisciplinary Perspectives*, 11, 100427. <https://www.mdpi.com/1660-4601/18/2/463>

Mayo, J. R., & Taboada, E. B. (2021). Commuter perceptions on the service quality of public transport in the Philippines: A case study of the PUV modernization program. *Journal of Public Transportation*, 23(1), 1-15. <https://doi.org/10.5038/2375-0901.23.1.2>

Montero-Gutiérrez, J., et al. (2023). Site-specific optimization of bus stop locations and climate-responsive shelter design. *Energy and Buildings*, 278, 112579.

https://www.researchgate.net/publication/360172864_Site-specific_optimization_of_bus_stop_locations_and_designs_over_a_corridor

National Center for Transportation Studies. (2024). Redesigning Philippine land transport governance towards improving commuter service quality. University of the Philippines Diliman.

National Economic and Development Authority. (2023). Philippine Development Plan 2023-2028: Chapter 12: Expand and Upgrade Infrastructure. Government of the Philippines.

Ong, A. K. S., Prasetyo, Y. T., Estefanio, A., Tan, A. S., Videña, J. C., Villanueva, R. A., Chuenyindee, T., Thana, K., Persada, S. F., & Nadlifatin, R. (2023). Determining factors affecting passenger satisfaction of “jeepney” in the Philippine urban areas: The role of service quality in sustainable urban transportation systems. *Sustainability*, 15(2), 1223.

<https://doi.org/10.3390/su15021223>

Paringit, E. C., Lucas, R. D., & Cutora, J. J. (2019). GIS for better public transportation and transit planning. *Philippine Journal of Science*, 148(4), 697–708.

<https://www.dlsu.edu.ph/wp-content/uploads/pdf/conferences/research-congress-proceedings/2019/see-I-004.pdf?>

Philippine News Agency. (2024, February 9). CDO opens 61 new routes under PUV modernization plan. <https://www.pna.gov.ph/articles/1218591>

Rahman, M. M., et al. (2020). Prioritizing metro service quality attributes using TOPSIS and importance-satisfaction analysis. *Journal of Public Transportation*, 23(1), 1–18.

<https://journals.sagepub.com/doi/10.1177/0361198120917972>

Ranosa, M. J., Fillone, A. M., & De Guzman, A. P. (2017). Jeepney service operation and demand analysis in Baguio City, Philippines. *Transportation Research Record*, 2650(1), 55–63.

<https://ncts.upd.edu.ph/tssp/wp-content/uploads/2017/07/TSSP2017-06-Ranosa-Fillone-and-De-Guzman.pdf>

Regan, A. C., et al. (2020). GIS-based identification and visualization of multimodal freight transportation catchment areas. *Transportation Research Part E*, 141, 102019.

<https://link.springer.com/article/10.1007/s11116-020-10155-3>

Regidor, J. R., & Aloc, R. C. (2016). Traffic problems at jeepney stops and proposals for improved jeepney stop policies. *Journal of the Eastern Asia Society for Transportation*

Studies, 11, 2135–2149.

<https://ncts.upd.edu.ph/tssp/wp-content/uploads/2018/08/Regidor96.pdf>

Reyes, J. M., et al. (2023). Moving toward objective metrics: The role of GPS telematics in evaluating modernized public utility vehicle performance. *Journal of Philippine Transportation Studies*, 15(2), 112–128.

Reyes, J. P., Cruz, R. A., & Santos, L. M. (2023). Measuring the efficiency of consolidated public transport routes using GPS data. *Journal of Transportation Engineering*, 149(6), 04023032.

Santos, T., Lobato, K., Rocha, J., & Tenedório, J. A. (2020). Modeling photovoltaic potential for bus shelters on a city scale: A case study in Lisbon. *Sustainable Cities and Society*, 60, 102255. <https://www.mdpi.com/2076-3417/10/14/4801>

Savilla, J. P., et al. (2024). Perception of jeepney drivers on the jeepney modernization program in Indang, Cavite. *Philippine Journal of Social Sciences*, 49(1), 55–68.

https://www.researchgate.net/publication/379494940_Perception_of_the_Jeepney_Drivers_on_the_Jeepney_Modernization_in_Indang_Cavite

Stadler, T., Hofmeister, S., & Dünneweber, J. (2022). A method for the optimized placement of bus stops based on Voronoi diagrams. *Proceedings of the 55th Hawaii International Conference on System Sciences (HICSS55)*. <https://doi.org/10.24251/HICSS.2022.694>

Ventura, A. B., & Regidor, J. R. (2018). Assessment of public utility vehicle modernization implementation: Data gaps and planning challenges. *Proceedings of the 25th Annual Conference of the Transportation Science Society of the Philippines*.

Ventura, A. B., Frayna, C. M., & Vergel, K. N. (2018). Determination of appropriate public transportation mode for a university transit route. *Philippine Engineering Journal*, 39(1), 45–58. <https://ncts.upd.edu.ph/tssp/wp-content/uploads/2018/07/TSSP2018-16.pdf>

Vergel, K. N., et al. (2014). Evaluation of compliance of customized local road vehicles with vehicle standards. *Philippine Engineering Journal*, 35(2), 1–14.

<https://ncts.upd.edu.ph/tssp/wp-content/uploads/2018/08/Vergel14.pdf>

Villegas, M. R., Ledres, J. M., & Estember, R. D. (2021). Physical assessment and perceived quality of e-jeepney in Metro Manila. *Journal of Transportation Technologies*, 11(3), 211–224. <https://www.ieomsociety.org/singapore2021/papers/1199.pdf>

Wang, F., Ye, M., Zhu, H., & Gu, D. (2022). Optimization method for conventional bus stop placement using Voronoi diagrams. *Journal of Advanced Transportation*, 2022, 1–12.

<https://www.mdpi.com/2071-1050/14/13/7918>

Wang, Y., Xu, Y., Wang, X., & Fan, X. (2017). Using GPS data to assess the operational performance of public bus routes. *Transportation Research Record: Journal of the Transportation Research Board*, 2652(1), 58–67.

Yigitcanlar, T., Sipe, N., Evans, R., & Pitot, M. (2007). Modeling pedestrian accessibility to public transport using GIS-based indices. *Environment and Planning B*, 34(5), 901–920.

Yu, Jingqiao & Zhu, Yuan & Wu, Chaowen & Song, Guoqing. (2024). Optimization of Bus Stop Location based on Spatial Analysis and K-Means Clustering Method – A Case Study in Hohhot City.

https://www.researchgate.net/publication/378528861_Optimization_of_Bus_Stop_Location_based_on_Spatial_Analysis_and_K-Means_Clustering_Method_-_A_Case_Study_in_Hohhot_City

Zhao, J., Copley, C., Wang, J., & Li, T. (2018). Analyzing the relationship between bus stop spacing and transit operational reliability. *Journal of Advanced Transportation*, 2018. Zulueta, M. C. (2024). Infrastructure and commuter experience of modernized public utility jeepneys using PLS-SEM. *Asian Transport Studies*, 8, 100112.

<https://www.mdpi.com/2071-1050/16/10/3912>

Zulueta, M. C., et al. (2024). Implications of the progression to sustainable public transportation: Insights into the modern jeepney shift. *Sustainability*, 16(5), 2247.

<https://www.mdpi.com/2071-1050/16/10/3912>

Saptutyningsih, E., & Karimah, N. (2019). Valuing public transportation using the contingent valuation method. *Economic Journal of Emerging Markets*.

<https://journal.umy.ac.id/index.php/esp/article/view/5017>

Fujii, S., et al. (2016). Knowledge, attitudes, and willingness to pay for sewerage and sanitation services in Metro Manila. *Journal of Environmental Science and Management*.

<https://ovcre.uplb.edu.ph/journals-uplb/index.php/JESAM/article/view/164>

APPENDIX A

Survey Questionnaire (English & Bisaya)

Instruction: Please check the box that corresponds to your answer or fill in the blanks provided.
(Instruksyon: Palihug butangi og takes ang kahon nga nag tugma sa imong tubag o sulati ang mga blanko nga gihatag.)

PART I: COMMUTER SOCIO- DEMOGRAPHIC PROFILE

[UNANG BAHIN: PROPILES SA SOCIO-DEMOGRAPHIC SA COMMUTER]

1. Age [Edad]:

- ☐ Below 18 *[Ubos sa 18 anyos]*
☐ 18 – 25 *[18- 25 anyos]*
☐ 26 – 35 *[26 - 35 anyos]*
☐ 36 – 45 *[36 - 45 anyos]*
☐ 46 and above *[46 anyos ug pataas]*

2. Occupation / Primary Activity [Trabaho / Pangunang Buluhaton]

- ☐ Student *[Estudyante]*
☐ Employee (Private Sector) *[Empleyado sa Pribadong Kompanya]*
☐ Government Employee *[Empleyado sa Gobyerno]*
☐ Self-employed / Business Owner / Vendor *[Adunay Kaugalingong Negosyo / Tindera/Tindero]*
☐ Unemployed / Homemaker *[Walay Trabaho / Nag Atiman sa Balay]*
☐ Other *[Uban pa]* (Please specify / Palihug isulatt): _____

PART II: ROUTE, ROAD CORRIDOR, AND TRAVEL BEHAVIOR (OBJECTIVE 1 & 3)

[IKADUHANG BAHIN: ROTA, KORIDOR SA DALAN, UG DETALYE SA PAGBIYAHE]

3. What specific Modern Jeepney Route / Cooperative Line are you riding today?
[Unsa nga saktong Rota sa Modern Jeepney o Cooperative Line ang imong ginasakan karon?]

Route Line *[Linya sa Rota]*: _____

4. Origin/Boarding Point (For k-means Clustering): Where do you usually get on this jeepney? *[Gikanan/Sakayan: Asa ka kasagaran mosakay niini nga jeepney?]*

Street Name / Landmark *[Dalan o Bantog nga Dapit]*: _____

Barangay: _____

5. Destination/Alighting Point (For k-means Clustering): Where do you usually get off

this jeepney? *[Padulngan/Naogan: Asa ka kasagaran manaog niini nga jeepney?]*

Street Name / Landmark *[Dalan o Bantog nga Dapit]*: _____

Barangay: _____

PART III: COMMUTER PERCEPTION ON FARE, DISTANCE, AND DELAY (Objective 2)

[IKATULONG BAHIN: PERSEPSYON SA PASAHE, GILAY-ON, UG PAGKALANGAN]

6. Fare [Pasahe]: How much is your modern jeepney fare for this specific route? *[Pila ang imong pamahe sa modern jeepney niini nga saktong rota?]*

- ☐ Minimum Fare (₱15.00) *[Plite nga minimum]*
☐ ₱16.00 – ₱25.00
☐ ₱26.00 – ₱35.00
☐ More than ₱35.00 *[Lapas ₱35.00]*

7. Distance [Gilay-on]: How long is the estimated distance of your single trip? *[Pila ang gibana-bana nga gilay-on sa imong usa ka biyahe?]*

- ☐ Short distance (Less than 4 km) *[Duol lang (Ubos sa 4 km)]*
☐ Medium distance (4 km – 8 km) *[Katamtaman (4 km hangtod 8 km)]*
☐ Long distance (More than 8 km) *[Layorason (Lapas sa 8 km)]*

8. Delay [Pagkalanngan]: On average, how much time delay do you experience during peak hours due to traffic congestion or waiting for boarding/unloading? *[Sa kasagaran, unsa ka dako nga pagkalanngan (delay) sa oras ang imong masinati tungod sa trapiko o sige'g hulat sa pag-sakay ug pag-naog?]*

- ☐ Minimal delay (0 – 5 minutes) *[Gamay nga langan (0 – 5 ka minuto)]*
☐ Moderate delay (6 – 15 minutes) *[Katamtaman nga langan (6 – 15 ka minuto)]*

- ☐ Heavy delay (16 – 30 minutes) [Dako nga langan (16 – 30 ka minuto)]
- ☐ Severe delay (More than 30 minutes) [Grabe nga langan (Lapas sa 30 ka minuto)]

9. Rate your level of satisfaction with the current modern jeepney service attributes: [I-rate ang imong lebel sa katagbawan sa mga mosunod nga aspeto sa modern jeepney:]
- (5 = Very Satisfied [Kontento Kaayo], 1 = Very Dissatisfied [Wala Kontento Kaayo])

Service Attribute [Aspeto sa Serbisyo]	5	4	3	2	1
Reasonable Fare Rate [Makiangayon nga Pasahe]					
Travel Time & Minimal Delays [Oras sa Biyahe ug Gamay nga Langan]					
Safety & Availability of Waiting Sheds [Kaluwasan ug Kakahimtang sa Waiting Shed]					

PART IV: CONTINGENT VALUATION & WILLINGNESS TO PAY (WTP) (OBJECTIVE 2 & 4)
[IKAUPAT NGA BAHIN: CONTINGENT VALUATION UG KAANDAMON MOBAYAD (WTP)]

Contingent Valuation Method (CVM) Scenario / [Sitwasyon sa Pag-assess]

English: Suppose the Local Government Unit (LGU) of Cagayan de Oro, in partnership with transport groups, establishes designated, well-lit modern passenger waiting sheds, loading zones, and terminals to minimize travel delays, resolve roadside chaos, and provide safety from extreme weather.

To guarantee the continuous cleanliness, structural maintenance, and security of these

commuter spaces, a localized transport facility maintenance program supported by a microscopic commuter contribution/fee per trip is being considered.

Bisaya: Ipapamatuod nato nga ang LGU sa Cagayan de Oro, inubanan sa mga transport group, magtukod og mga moderno ug hayag nga waiting sheds, loading zones, ug terminals aron mamenosan ang pagkalangan (delay) sa biyahe, han-ayon ang sakayan, ug maghatag og proteksyon sa init ug ulan.

Aron masiguro ang padayon nga kalimpyo, pagmentinar, ug seguridad niini nga mga dapit, adunay gikonsidera nga gamay nga amutan o commuter fee kada biyahe para sa pagmentinar sa pasilidad.

10. Given this scenario, are you willing to contribute a nominal micro-fee to support the sustained installation, operations, and strict maintenance of these modern commuter facilities? [Subay niini, andam ka ba nga mo-amot og gamay nga kantidad aron suportahan ang padayon nga pagpadagan ug pagmentinar niini nga mga moderno nga waiting sheds ug terminals?]
- ☐ Yes (Proceed to Question 11) / [Oo (Padayon sa Pangutana 11)]
- ☐ No (End of Survey) / [Dili (Katapusan sa Survey)]

11. If YES, what is the maximum amount you are willing to contribute as a micro-fee per trip to ensure these facility locations stay functional, safe, and clean? [Kon Oo, unsa ang pinakadako nga kantidad nga andam nimong i-amot isip micro-fee kada biyahe aron masiguro nga magpabilin nga hapsay, luwas, ug limpyo kini nga mga pasilidad?]
- ☐ ₱2.00
- ☐ ₱5.00
- ☐ ₱10.00
- ☐ More than ₱10.00 [Lapas sa ₱10.00]

APPENDIX B

Informed Consent Form (English & Bisaya)

INFORMED CONSENT FORM

Project Title: **GIS-Based Analysis of Commuters Facility and Jeepney Routes Using Contingency Valuation and K-means Clustering along Barangay Gusa to Barangay Bugo in Cagayan de Oro City.**

Principal Researcher: Kacey Capiña

Contact Number: 09368212308

Email: 20200019356@my.xu.edu.ph

Department of Civil Engineering, Xavier University - Ateneo de Cagayan

This is a survey questionnaire conducted by Xavier University, Civil Engineering Department of the College of Engineering. The research study is entitled **"GIS-Based Analysis of Commuters Facility and Modern Jeepney Routes Using Contingency Valuation and K-means Clustering in Cagayan de Oro City."** The main objective of the study is to evaluate commuter conditions and prioritize locations for passenger facilities for Modern Jeepneys. Specifically, the study aims to:

1. Identify the baseline road corridor and existing routes of modern jeepneys.
2. Determine commuter travel attributes such as fare, distance, and delay using the Contingent Valuation Method (CVM) and Willingness to Pay (WTP).
3. Evaluate the spatial distribution of commuters to identify high-demand boarding/alighting areas using the K-means clustering algorithm.
4. Recommend optimal locations for passenger waiting sheds and modern jeepney terminals using Geographic Information Systems (GIS).

This consent form is necessary for us to ensure that you understand the purpose of your involvement and that you agree to the conditions of your participation. This survey would take only about 10–15 minutes. You will be asked about your background, including basic demographic information, your specific boarding/alighting locations, your perceptions regarding travel attributes (fares, distances, delays), and your willingness to contribute a micro-fee to support the strict maintenance and security of improved commuter facilities. Your responses will be used solely for research purposes and will remain confidential.

Bisaya:

Kini usa sa ka pamanutana (survey questionnaire) nga gipahigayon sa Xavier University, Civil Engineering Department ubos sa College of Engineering. Ang titulo niini nga pagtulon-an mao ang "GIS-Based Analysis of Commuters Facility and Modern Jeepney Routes Using Contingency Valuation and K-means Clustering in Cagayan de Oro City."

Ang nag-unang katuyoan niini nga research mao ang pag-timbang-timbang sa kahimtang sa mga commuter ug pag-ila sa mga prayoridad nga lokasyon alang sa mga waiting shed ug terminal sa Modern Jeepney. Sa saktong pagkasulti, kini nga pagtulon-an nagtumong sa:

1. *Pag-ila sa baseline nga koridor sa dalan ug mga kasamtangang rota sa modern jeepney.*
2. *Pagkuha sa mga detalye sa biyahe sa commuter sama sa pasahe, gilay-on, ug pagkalangan (delay) gamit ang Contingent Valuation Method (CVM) ug Willingness to Pay (WTP).*
3. *Pag-assess sa spatial distribution o dapit sa pagdagsang sa mga commuter aron mailhan ang mga dapit nga taas ang demand sa sakayan ug naogan gamit ang k-means clustering algorithm.*
4. *Paghatag og rekomendasyon alang sa labing maayong lokasyon sa mga waiting shed ug modern jeepney terminal gamit ang Geographic Information Systems (GIS).*

Kini nga porma sa pasugot gikinahanglan aron masiguro namo nga imong nasabtan ang katuyoan sa imong pag-apil ug nga ikaw miuyon sa mga kondisyon niini. Kini nga survey mokabat lang og 10–15 ka minuto. Pangutan-on ka bahin sa imong profile (demographics), ang imong saktong sakayan ug naogan, ang imong opinyon bahin sa pamahe, gilay-on, ug pagkalangan (delays), ug ang imong kaandamon nga mo-amot og gamay nga micro-fee alang sa

pagmentinar ug seguridad sa mga modernong waiting sheds. Ang tanan nimong tubag gamiton lamang alang sa research ug magpabilin nga kompidensyal.

Statement of Consent / Propesiya sa Pasugot

English:

- I understand the purpose of this study and what is required of me.
- I understand that my participation is completely voluntary and that I can choose to withdraw or stop answering at any time without any penalty.
- I understand that all data gathered will be treated with strict confidentiality in compliance with the Data Privacy Act of 2012.
- By filling out and submitting this questionnaire, I voluntarily agree to participate in this study.

Bisaya:

- *Nasabtan nako ang katuyoan niini nga pagtulon-an ug unsa ang gihangyo kanako.*
- *Nasabtan nako nga ang akong pag-apil hingpit nga boluntaryo ug mahimo nako hunongon ang pagtubag bisan unsang orasa nga walay silot.*
- *Nasabtan nako nga ang tanang data nga makuha magpabilin nga pribado ug protektado subay sa Data Privacy Act of 2012.*
- *Pinaagi sa pagsulat ug paghatag niini nga pamanutana, boluntaryo akong miuyon nga moapil niini nga pagtulon-an.*

Name and Signature of Participant: _____

Date: _____

APPENDIX C

Curriculum Vitae

KACEY CAPIÑA

SILVERCREEK SUBDIVISION, PHASE 1, BLOCK 15 LOT 9,
CARMEN, CAGAYAN DE ORO CITY
capinakacey@gmail.com
0936-821-2308



EDUCATIONAL ATTAINMENT

Bachelor of Science in Civil Engineering Xavier University – Ateneo de Cagayan	August 2024	-
73 Corrales Ave, Cagayan de Oro City, 9000 Misamis Oriental	Present	
University of Science and Technology of Southern Philippines	August 2022	-
Claro M. Recto Ave, Lapasan, Cagayan de Oro City, 9000 Misamis Oriental	January 2024	
Senior High School – Humanities and Social Sciences		
Xavier Ateneo Senior High School	June 2020	-
Fr. William Masterson SJ Ave, Upper Balulang, Cagayan de Oro City, 9000 Misamis Oriental	April 2022	
Junior High School - Misamis Oriental General Comprehensive Highschool (MOGCHS)		
Don Apolinar Velez St, Cagayan de Oro City, 9000 Misamis Oriental	June 2016	to March 2020
Elementary - City Central School		
P.N. Roa St, Brgy. 29, Cagayan de Oro City, 9000 Misamis Oriental	June 2013	- March 2016
Preschool to Elementary - St. Mary's Academy of Carmen		
Villarin St, Carmen, Cagayan de Oro City, 9000 Misamis Oriental	June 2009	- March 2013

SKILLS

- Knowledgeable in Computer Aided Design (AutoCAD, Sketchup)
- Knowledgeable in Mapping (ArcGIS, QGIS, Google Earth Pro)
- Knowledgeable in Microsoft Office softwares (Excel, Word, PowerPoint)
- Visual Communication (Canva & Photoshop for Graphic Design, Capcut for Video Editing)
- Public Speaking Skills
- Communication Skills
- Organizational Governance Skills
- Event Coordination Skills
- Technical Writing Skills
- Budgeting and Resource Management

RELATED EXPERIENCE/S

POSITION	COMPANY NAME	COMPANY ADDRESS	INCLUSIVE DATE
Creative Assistant	Virtual Kacey Capiña	Silvercreek Subdivision	January 2023 - December 2023

Volunteer for Secretariat Philippine Works (under a Police Captain Officer)	National Claro M. Recto Ave, Lapasan, Oro City, 9000 Misamis Oriental	April 2024 - June 2024
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Representative (for a National Conference Engineering Students (USPICE) - CDO 2025)	Unified Students of Cagayan de Oro City	November 24 - 25, 2025
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AWARDS

NAME/TYPE OF AWARD	AWARD-GIVING BODY	DATE AWARDED
Consistent Honor Student	St. Mary's Academy of Carmen	June 2009 – March 2013
Consistent Honor Student	City Central School	June 2019 – July 2023
Consistent Honor Student	MOGCHS	June 2016 - March 2020
Second Honors	Xavier Ateneo Senior High School	June 2021
Best Paper Category)	(HUMSS Xavier Ateneo Senior High School	May 2022

ORGANIZATIONS

POSITION	NAME OF ORGANIZATION	INCLUSIVE DATES
Associate Head	Department of Students' Health Development; Graphic Design	July 2021 – April 2022
Member	Leadership Empowerment and Development Society - USTP ; Committee for Multimedia Production	August 2022 - June 2023
Associate Head	Philippine Institute of Civil Engineers – Xavier University Student Chapter (PICE-XUSC); InPro Committee	August 2024 – July 2025
Shadow Minister	Prime Philippine Institute of Civil Engineers – Xavier University Student Chapter (PICE-XUSC); House of Representatives	August 2025 – Present

Assistant	Unified Students of Philippine Institute of Civil	September 2025 –
Treasurer	Engineers (USPICE); Executive Officers	Present

References (available upon request).

Data Privacy Consent

In compliance with the Data Privacy Act (DPA) of 2012, and its implementing Rules and Regulations (IRR) effective September 8, 2016,

I allow ☒ (I hold free and harmless and indemnify Xavier University from any complaint, suit or damages which any party may file or claim in relation to my consent)

I don't allow ☐ (My resume will be used solely for counseling purposes)


Signature of student _____

Signed this 29th day of April 2026 at Cagayan de Oro City.

CAROLINE PHILOMENA CUNDANGAN

Blk 15, Lot 26, Bellevue Subdivision, Upper Carmen, CDOC

cpccundangan@gmail.com

09150978816



EDUCATIONAL ATTAINMENT

Bachelor of Science in Civil Engineering

Xavier University – Ateneo de Cagayan Corrales Ave., Cagayan de August 2023 - Present
Oro City

Senior High School – Science, Technology, Engineering and

Mathematics

October 2020 – June

Pagadian Science High School

2022

Rizal Ave, Lumbia, Pagadian City, Zamboanga del Sur

Junior High School - Saint Columban College Junior High School

Enerio Street, Pagadian City, Zamboanga del Sur

June 2017 – March

2020

Elementary - Pagadian City Pilot School

San Jose District, Pagadian City, Zamboanga

June 2010 - March

Del Sur

2016

SKILLS

- Basic Computer Aided Design (CAD; AutoCAD, Sketchup)
- Basic Mapping Skills (ArcGIS, QGIS, Google Earth Pro)
- Basic Programming Skills (Python)
- Knowledgeable in Microsoft Office Applications (Word, Excel, Powerpoint, Etc.)
- Solid Communication Skills
- Time Management Skills

RELATED EXPERIENCE/S

POSITION	COMPANY NAME	COMPANY ADDRESS	INCLUSIVE DATE
Volunteer	Saint Columban College Junior High School	- Enerio Street, Pagadian City, Zamboanga del Sur	June 2018 – May 2020
Volunteer	Pagadian Science High School	Rizal Ave, Lumbia, Pagadian City, Zamboanga del Sur	April 2022
ACES – Events Committee Member	Xavier University - Association of the College of Engineering Students	Corrales Ave., Cagayan de Oro City	September 2023

AWARDS

NAME/TYPE OF AWARD	AWARD-GIVING BODY	DATE AWARDED
Honor Student	Pagadian City Pilot School	March 2014 - March 2016
Consistent Honor Student	Pagadian Science High School	July 2021 - June 2022
Best Paper (Stem Category)	Pagadian Science High School	July 2023

ORGANIZATIONS

POSITION	NAME OF ORGANIZATION	INCLUSIVE DATES
Math Cub Member	Saint Columban College Junior High School	August 2017– March 2020

Science Club Member Saint Columban College Junior High School August 2017 - March 2020

Youth Member Philippine Red Cross - Zamboanga del Sur October 2018 - June 2022
Chapter

Angklung Ensemble Saint Columban College Junior High School September 2018 – March 2020
Member

ACES – Events Xavier University - Association of the September 2023
Committee Member College of Engineering Students

References (available upon request).

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Signature of student  _____

Signed this 1st day of May 2026 at Cagayan de Oro City.

BEA ARABELLA LOPEZ

Phase 3A, Villa Trinitas Subd., Bugo, Cagayan de Oro City

lopez.beaarabella@gmail.com

09939759721



EDUCATIONAL ATTAINMENT

Bachelor of Science in Civil Engineering	Xavier University – Ateneo de	
Cagayan Corrales Ave., Cagayan de Oro City		August 2023 - Present
Senior High School – Science, Technology, Engineering and Mathematics		
Liceo de Cagayan University		August 2021 –
Kauswagan, Cagayan de Oro City		April 2023
Junior High School		
Bugo National High School		June 2017 –
Rabocca drive, Bugo, Cagayan de Oro City		April 2021
Elementary		
Bugo Central School		June 2015 –
Bantilles, Bugo, Cagayan de Oro City		April 2017
Elementary		
Philippine Southfield School		June 2001–
Phs 1, Villa Trinitas Subd., Bugo, Cagayan de Oro City		March 2015

SKILLS

- Knowledgeable in Computer Aided Design (AutoCAD/Sketchup)
- Proficient in Mapping (QGIS, ArcGIS)
- Proficient in Microsoft Office softwares (Word, Excel, PowerPoint)
- Quantity Take-Off and Cost Estimation
- Adept Interpersonal and Leadership Skills
- Flexible and can work under pressure
- Safety Awareness in Construction Sites
- Time Management skills
- Visual Communication (Canva for Graphic Design, Capcut for Video Editing)
- Strong Analytical and Problem-Solving Skills
- Ability to Work Independently and in a Team
- Attention to Detail and Accuracy

RELATED EXPERIENCE/S

POSITION	COMPANY NAME			COMPANY ADDRESS	INCLUSIVE DATE
Volunteer	Parish Council	Youth Coordinating	Bugo, Cagayan	Oro City	de August 2021 – December 2023
Volunteer	Bugo National High School			Bugo, Cagayan	de July 2018 – May 2021
				Oro City	
Volunteer	Parish Pastoral Council	Responsible Voting	for Bugo, Cagayan	Oro City	de May 10, 2022

AWARDS

NAME/TYPE AWARD	OF AWARD-GIVING BODY	DATE AWARDED
Consistent Achiever	Philippine Southfield School	June 2011 – May 2015
Consistent Honor Student	Bugo Central School	June 2015 – May 2017
Consistent Honor Student	Bugo National High School	June 2019 – May 2021

ORGANIZATIONS

POSITION	NAME OF ORGANIZATION	INCLUSIVE DATES
Ministry Head – Events and Presentations	Philippine Institute of Civil Engineers – Xavier University Student Chapter	August 2025 – May 2026
Associate Ministry Head – Publication Communication	Philippine Institute of Civil Engineers – Xavier University Student Chapter	October 2024 – May 2025
Member	Motion Ensemble	September 2022 – May 2023

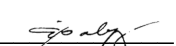
References (available upon request).

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I don't allow ☐ (My resume will be used solely for counseling purposes)

Signature of student 

Signed this 30th day of April 2026 at Cagayan de Oro City.

JAMMEZA USMAN

Camella Homes, Prime Rose St. Block 15 Lot 59-61, Upper
Balulang, Cagayan de Oro
jammezausman@gmail.com
0955-462-1121



EDUCATIONAL ATTAINMENT

Bachelor of Science in Civil Engineering Xavier University – Ateneo de Cagayan Corrales Ave., Cagayan de Oro City	August 2023 - Present
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Senior High School – Science, Technology, Engineering, and Mathematics Xavier Ateneo Senior High School	June 2021 -
Fr. William Masterson SJ Ave, Upper Balulang, Cagayan de Oro City, April 2023 9000 Misamis Oriental	

Junior High School – Corpus Christi School, Masterson Avenue, Cagayan de Oro City, 9000 Misamis Oriental.	June 2017 - April 2021
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Elementary - Corpus Christi School, Masterson Avenue, Cagayan de Oro City, 9000 Misamis Oriental.	June 2011 - March 2017
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SKILLS

- Knowledgeable in Computer Aided Design (AutoCAD, Sketchup)
- Knowledgeable in Mapping (ArcGIS, QGIS, Google Earth Pro)
- Knowledgeable in Structural Analysis (Frame Analysis, Vertical Reactions, Earthworks Volume)
- Knowledgeable in Construction Estimation (NSCP Standards, Formworks, Concrete Volume)
- Knowledgeable in Microsoft Office softwares (Excel, Word, PowerPoint)
- Visual Communication (Canva & Photoshop for Graphic Design, Capcut for Video Editing)
- Communication Skills
- Organizational Governance Skills
- Technical Writing Skills
- Critical Thinking and Problem Solving

RELATED EXPERIENCE/S

POSITION	COMPANY NAME	COMPANY ADDRESS	INCLUSIVE DATE
Volunteer	Corpus School	Christi Masterson Avenue, Cagayan de Oro City, 9000 Misamis Oriental.	June 2011 – April 2017
Academic Project Lead	Corpus School	Christi Masterson Avenue, Cagayan de Oro City, 9000 Misamis Oriental.	July 2022 – August 2022
Student Leader	Corpus School	Christi Masterson Avenue, Cagayan de Oro City, 9000 Misamis Oriental.	June 2017 – April 2021

AWARDS

NAME/TYPE OF AWARD	AWARD-GIVING BODY	DATE AWARDED
Academic Excellence in Mathematics (Best in Math)	Corpus Christi School	Christi March 2018, March 2019, April 2021
Consistent Honor Student	Corpus Christi School	Christi June 2017 – April 2021

ORGANIZATIONS

POSITION	NAME OF ORGANIZATION	INCLUSIVE DATES
Student Council	Corpus Christi School	June 2017 – April 2021
Mathematics Society Member	Corpus Christi School	June 2011 – April 2021
Computer Club Member	Corpus Christi School	June 2017 – April 2021
Science Club Member	Corpus Christi School	June 2015 – April 2021
Sports Organization Member	Corpus Christi School	June 2014 – March 2020
Associate Head	Philippine Institute of Civil Engineers – Xavier University Student Chapter (PICE-XUSC); Pubcom Committee	August 2023 - Present
Media and Arts Member	Unified Students of Philippine Institute of Civil Engineers (USPICE); Executive Officers	July 2025 - December 2025

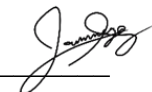
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Signature of student 

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